

Experiment Report of Lab 1-4

Name: 王泳淇 ID: 3180105481 Major: Computer Science (Pursuit Science Class)

Course: Logic and Computer Design Fundamentals Group mates: 朱允怀、沈栩杰等

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Lab 1 Introduction to Common Electronic Instruments

1. Purposes and Requirements of the Experiment

- 1.1 To learn about common electronic instruments.
- 1.2 To learn how to use digital oscilloscope, digital signal generator (function signal generator), DC stabilized power supply, multimeter and other electronic instruments.
- 1.3 To master how to measure the waveform, amplitude and frequency of an impulse with a digital oscilloscope.
- 1.4 To master how to measure the rising and falling edges, delay and other parameters of a pulse timing with a digital oscilloscope.
- 1.5 To master how to measure voltage, resistance and to judge the state of a diode with a multimeter.

2. Tasks and Principles of the Experiment

2.1 Tasks

1. Use the digital oscilloscope to measure the parameters of the wave generated by the function signal generator. Switch the frequency of the signal to 100Hz, 10kHz and 100kHz and the waveform to sinusoidal wave. Read the period and frequency of the signal with the oscilloscope, and check the accuracy of the signal generator.
2. Switch the frequency of the sinusoidal wave generated by the generator to 1kHz(50Hz) and the peak-peak amplitude to 4-6V. (Use AC gear of the multimeter to measure the

rms value.) Then use the oscilloscope to measure the amplitude of the wave. Compute and compare the rms voltages.

2.2 Principles

1. Digital oscilloscope is an electronic instrument that can read digital signals and display the curve of the signal. It is often used to measure the parameters of digital signals.
2. Functional signal generator can generate different kinds of digital signals. It's often used as a supplement of oscilloscopes.
3. Multimeter is a tool to measure current, voltage and resistance. It can also tell the polarity of diodes and triodes.

3. Mainly-used Instruments and Materials

Instruments

- | | |
|--|---|
| 1. digital oscilloscope RIGOL-DS162 | 1 |
| 2. function signal generator YB1638 | 1 |
| 3. digital multimeter | 1 |
| 4. experiment box of logic circuit designing | 1 |

Materials

- | | |
|---------------------------------|---------|
| 5. common electronic components | several |
|---------------------------------|---------|

4. Procedures and Operations of the Experiment

4.1 Measure the parameters of sinusoidal wave with the oscilloscope.

Firstly, adjust the YB1638 signal generator by the frequency range switch and the frequency adjusting knob so that it can generate sinusoidal wave of 100Hz, 10kHz and 100kHz. We can read the frequency from the numeric display of the generator. Connect the output with Channel 1 of the oscilloscope and connect the ground clamps together.

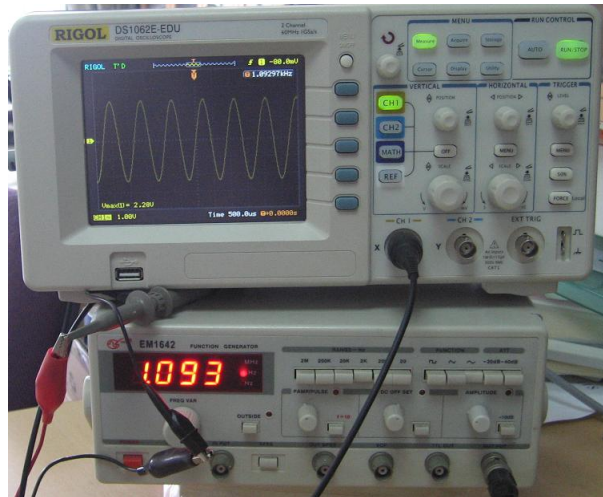


Figure 1 Signal generator and oscilloscope

Then measure the voltage, period and frequency of the signal with the oscilloscope. Compare the data with the output of the generator, and fill the table below.

Figure 2 Measuring the parameters of sinusoidal wave

	Generator output	Oscilloscope reading	Sensitivity	Measured Value (Peak-peak)	
p-p value	3V~5V	Div	V/Div	V	
Period/Frequency	100Hz	Div	ms/Div	ms	Hz
p-p value	2V~10V	Div	V/Div	V	
Period/Frequency	10kHz	Div	μ s/Div	μ s	kHz
p-p value	4V~6V	Div	V/Div	V	
Period/Frequency	100kHz	Div	μ s/Div	μ s	kHz

4.2 Measure the output voltage of YB1638 signal generator.

Switch the frequency to 1kHz and the peak-peak voltage to 4V-6V. Connect the output with the oscilloscope and measure the amplitude. Then use the AC gear of the multimeter to measure the amplitude. Fill the table and compare the rms value computed from the oscilloscope readings and the multimeter readings.



Figure 2 Measuring voltage with multimeter

Figure 3 Measuring the output voltage

Generator output frequency	Oscilloscope reading		Computed RMS value	Multimeter reading
1kHz	Div	V/Div	V	V
50Hz	Div	V/Div	V	V
1mHz (not accurate)	Div	V/Div	V	V

4.3 Measure the DC voltage of the experiment box with a multimeter

Plug the red probe into the VΩmA socket and the black probe into the COM socket. Switch to the DC gear of the multimeter, and connect the probes to the circuit. The polarity of the end connected to the red probe will be displayed on the numeric display. At the same time, use the oscilloscope to measure the output of the DC stabilized power supply. Record and compare the readings.

Figure 4 Measuring the DC voltage

DC stabilized power supply output	Oscilloscope reading	Sensitivity	Computed value of oscilloscope reading	Multimeter reading
+5V	Div	V/Div	V	V

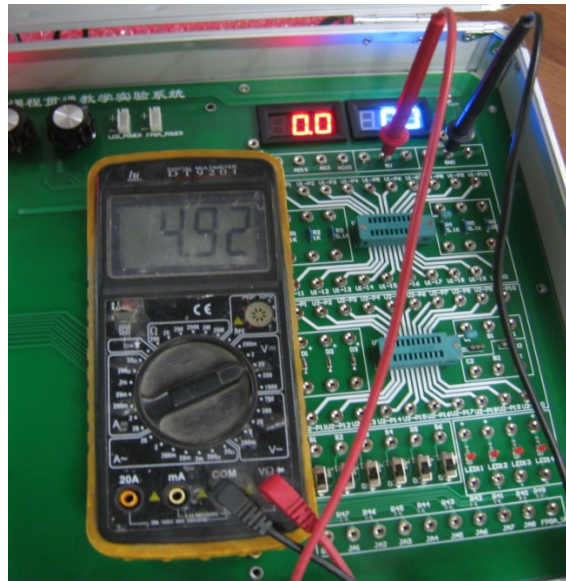


Figure 5 Measuring the DC voltage with the multimeter

4.4 Test the state of diode with a multimeter

Plug the red probe into V Ω socket and the black probe into COM socket. Switch the multimeter to ∇ (diode) gear. Connect the probes to the poles of the diode. If the reading is between 0.6 and 0.7, it means the red probe is connected to the anode. Otherwise if the reading is 1., it means the red probe is connected to the cathode.

5. Analysis and Result of the Experiment

5.1 Measure the parameters of sinusoidal wave with the oscilloscope.

Figure 6 Measuring the parameters of sinusoidal wave with the oscilloscope

	Generator output	Oscilloscope reading	Sensitivity	Measured Value (Peak-peak)	
p-p value	5V	4.8Div	1V/Div	4.8V	
Period/Frequency	100Hz	4.7Div	2ms/Div	9.4ms	106.4Hz
p-p value	5V	4.8Div	1V/Div	4.8V	
Period/Frequency	10kHz	4.6Div	20 μ s/Div	92 μ s	10.9kHz
p-p value	5V	4.9Div	1V/Div	4.9V	
Period/Frequency	100kHz	5.2Div	2 μ s/Div	10.4 μ s	96.1kHz

Analysis: We can see from the table that the voltage and frequency measured by the oscilloscope are generally consistent with the reading of the generator, but there is still an

error between them. This error comes from several ways. The output of generator is unstable and does not always match with its reading. The oscilloscope has an instrumental error and there is a voltage loss. Besides, a visual error is generated while reading the scale of the oscilloscope.

5.2 Measure the output voltage of YB1638 signal generator.

Figure 7 Measuring the output voltage of YB1638 signal generator

Generator output frequency	Oscilloscope reading		Computed RMS value	Multimeter reading
1kHz	4.8Div	1V/Div	1.69V	1.638V
50Hz	4.8Div	1V/Div	1.69V	1.641V
1mHz (not accurate)	4.8Div	1V/Div	1.69V	0.002V

Analysis: When the frequency is low, the reading of the oscilloscope is generally consistent with the reading of the multimeter. There is only a little instrumental and visual error. But when the frequency is about 1mHz, the multimeter is not able to measure the output voltage correctly. This is because multimeters are calibrated in a certain range of frequency. Their inner circuits cannot sample the high-frequency signal correctly, and thus cannot get the correct voltage value.

5.3 Measure the DC voltage of the experiment box with a multimeter

Figure 8 Measuring the DC voltage of the experiment box with a multimeter

DC stabilized power supply output	Oscilloscope reading	Sensitivity	Computed value of oscilloscope reading	Multimeter reading
+5V	5.5Div	1V/Div	5.5V	5.00V

Analysis: It can be seen that compared to the oscilloscope, the multimeter has a higher accuracy. This is because digital multimeter is a rather precise instrument and has less instrumental error while oscilloscope has more instrumental and visual error.

5.4 Test the state of the diode with a multimeter

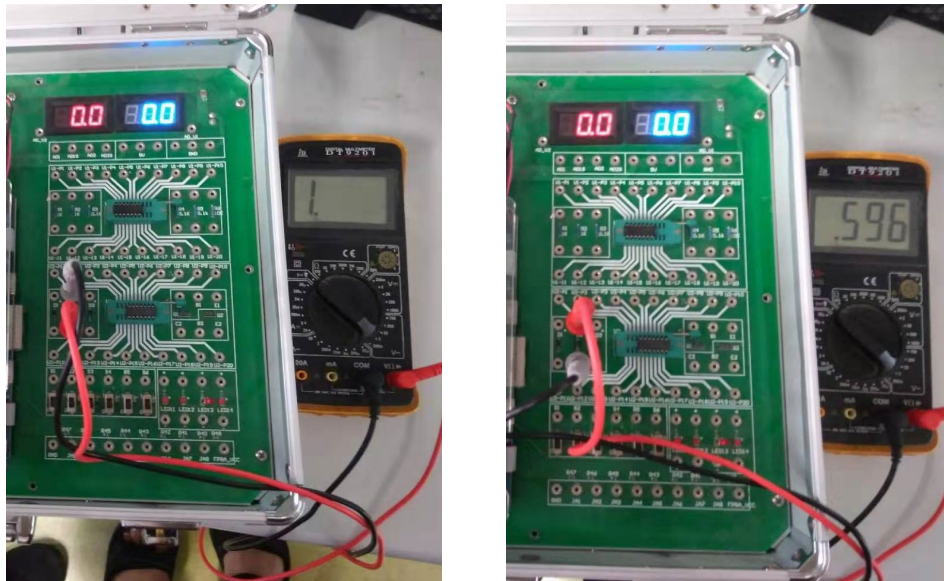


Figure 9 Testing the state of diode (left: reverse right: forward)

Analysis: When the multimeter is switched to ∇ (diode) gear, it displays the voltage drop on the diode. In the left picture, the voltage drop is over 1V. This means the red probe is now connected to the cathode of the diode. In the right picture the voltage drop is about 0.596V, which means the red probe is connected to the anode.

6. Discussion and Revision of the Experiment

The first experiment is rather simple to do and it didn't take me much time. Even so, I feel that this experiment is of great significance. It's the basis of the following series of experiments. For example, in the following experiments, we will need to measure the electric attributes of electronic instruments like logic gate circuits. Without multimeters and oscilloscopes, we won't be able to do this. And semiconductors like diodes are the core of logic circuits gates, only if we fully comprehend their characteristics can we understand the functions of logic gates.

Through this experiment, I get more familiar with the electronic instruments and can now use them expertly. Moreover, by cooperating with classmates to measure and record parameters, I can now better collaborate with others in doing experiments.

Lab 2 Basic Switching Circuit

1. Purposes and Requirements of the Experiment

- 1.1 To master the basic structure of a logical circuit.
- 1.2 To learn about the concepts of turn-on and cut-off of a diode.
- 1.3 To learn to construct simple logic gates with diodes and triodes.
- 1.4 To learn about the structures of the simplest logic gates.

2. Tasks and Principles of the Experiment

2.1 Tasks

1. Assemble an AND gate and a OR gate with diodes, and a NOT gate with triodes, then measure their input and output voltages and analyze their logical functions.
2. Assemble a NAND gate with the AND gate and the NOT gate. Measure its input and output voltages and analyze its logical function.

2.2 Principles

Diode is a kind of semiconductor elements. It only allows current to flow from its anode to cathode. It outputs different logic values depending on the input electrical level. For this reason, diodes can be used in logic gate circuits. Here is the table of its I/O characteristics.

Figure 10 I/O characteristics of semiconductor instruments

Logic voltage	V_{CC} / V	V_{OH} / V	V_{OL} / V	V_{IH} / V	V_{IL} / V	Description
TTL	5	≥ 2.4	≤ 0.4	≥ 2.0	≤ 0.8	Floating input is high level
LVTTL	3.3	≥ 2.4	≤ 0.4	≥ 2.0	≤ 0.8	
LVTTL	2.5	≥ 2.0	≤ 0.2	≥ 1.7	≤ 0.7	
COMS	5	≥ 4.45	≤ 0.5	≥ 3.5	≤ 1.5	Input resistor is big
LVCOMS	3.3	≥ 3.2	≤ 0.1	≥ 2.0	≤ 0.7	
LVCOMS	2.5	≥ 2.0	≤ 0.1	≥ 1.7	≤ 0.7	
RS232	$\pm 12 \sim 15$	$-3 \sim -5$	$3 \sim 15$	$-3 \sim -15$	$3 \sim 15$	Negative logic

Only TTL and COMS are needed in this experiment.

Triode is another kind of semiconductor element. It has three poles: base, collector and

emitter. It can also be used in logic gate circuits.

3. Mainly-used Instruments and Materials

Instruments

1. digital oscilloscope RIGOL-DS162	1
2. function signal generator YB1638	1
3. digital multimeter	1
4. experiment box of logic circuit designing	1

Materials

5. diodes of type IN4001	3
6. triode of type 9013	1
7. light emitting diode	1
8. resistances	
8.1 10k Ω	5
8.2 5.1k Ω	3
8.3 1k Ω	5

4. Procedures and Operations of the Experiment

4.1 Assemble an AND gate with diodes

1. Connect the circuits according to the diagram and check whether the diodes, power supply and resistances are in working condition.
2. Connect V_{CC} to the +5V socket in the experiment box.
3. The high and low input levels are generated by switches S1-S6. Input different level combinations of A and B. Measure the voltages V_A , V_B and corresponding output voltage V_F . Record them into the table, and judge if the logic relation satisfies $F = AB$.

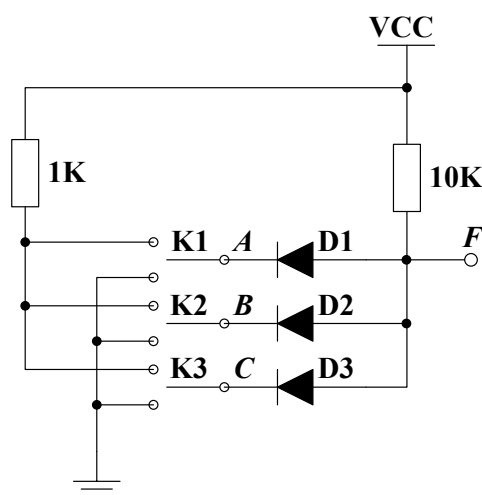


Figure 11 AND gate

Figure 12 I/O level of AND gate

V_A / V	V_B / V	V_F / V	Logic value of F

4.2 Assemble an OR gate with diodes

1. Connect the circuits according to the diagram and check whether the diodes, power supply and resistances are in working condition.
2. Connect V_{CC} to the +5V socket in the experiment box.
3. The high and low input levels are generated by switches S1-S6. Input different level combinations of A and B. Measure the voltages V_A , V_B and corresponding output voltage V_F . Record them into the table, and judge whether the logic relation satisfies $F = A + B$.

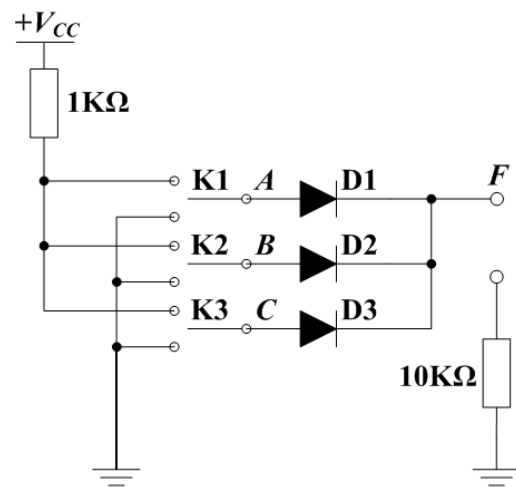


Figure 13 OR gate

Figure 14 I/O level of OR gate

V_A / V	V_B / V	V_F / V	Logic value of F

4.3 Assemble a NOT gate with a Triode.

1. Connect the circuits according to the diagram

and check whether the triode, power supply and resistances are in working condition.

2. Connect V_{CC} to the +5V socket in the experiment box.
3. The high and low levels inputted to A are generated by switches S1-S6. Measure the input voltage V_A and the corresponding output V_F . Record the data into the table, and justify whether the logic relation satisfies $F = \bar{A}$.

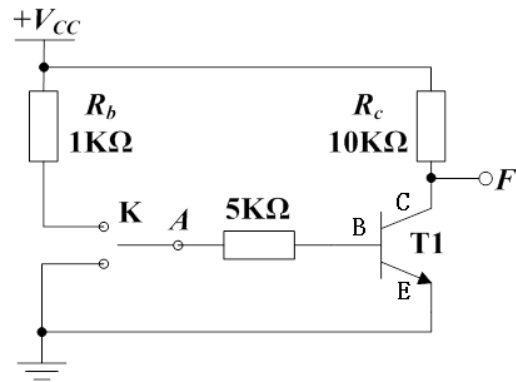


Figure 15 NOT gate

Figure 16 I/O level of NOT gate

V_A / V	V_F / V	Logic value of F

4.4 Assemble a NAND gate with transistors.

1. Connect the circuits according to the diagram and check whether the transistors, power supply and resistances are in working condition.
2. Connect V_{CC} to the +5V socket in the experiment box.
3. The high and low levels inputted to A and B are generated by switches S1-S6. Measure the input voltages V_A , V_B and corresponding output voltage V_F . Record them into the table, and judge whether the logic relation satisfies $F = \overline{AB}$.

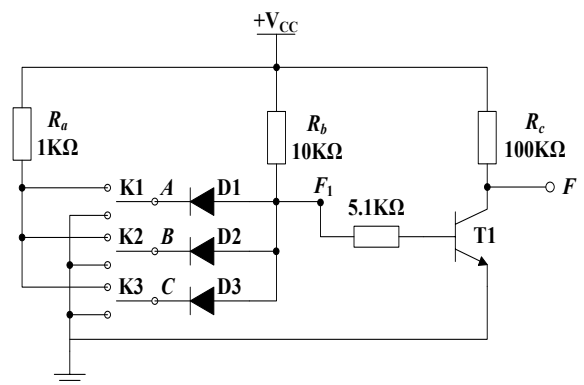


Figure 17 NAND gate

Figure 18 I/O level of NAND gate

V_A / V	V_B / V	V_F / V	Logic value of F

5. Analysis and Result of the Experiment

5.1 Assemble an AND gate with diodes

The following 4 pictures show 4 states of AND gate corresponding to the states in the table.

The red numerical reading shows V_A while the blue one shows V_F . Since V_A and V_B are symmetric, we can assume that the high and low level of V_B is the same as V_A .

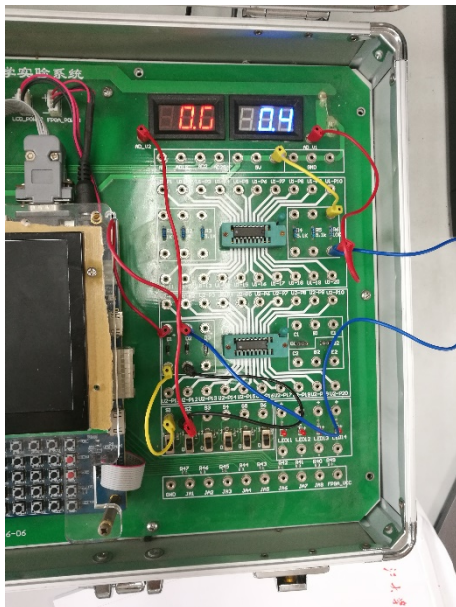


Figure 19 AND gate (A: low B: low)

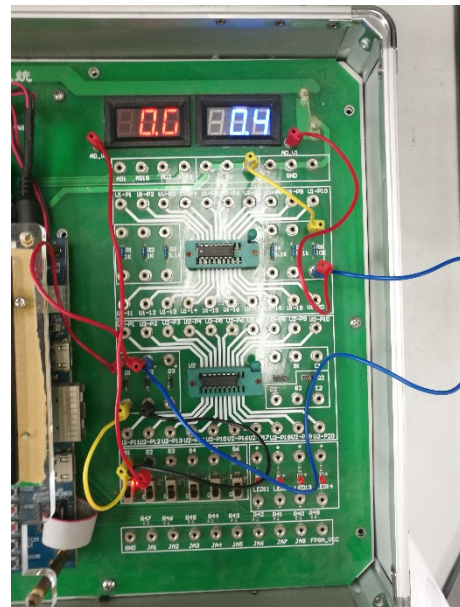


Figure 20 AND gate (A: low B: high)

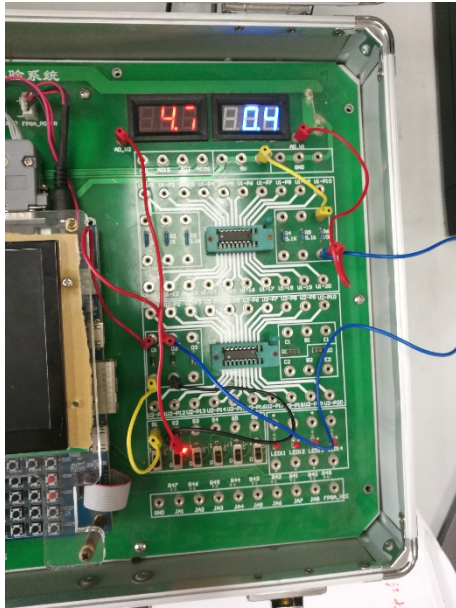


Figure 21 AND gate (A: high B: low)

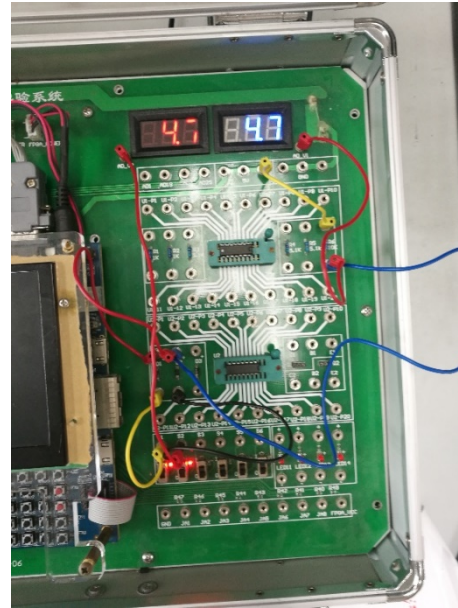


Figure 22 AND gate (A: high B: high)

Figure 23 I/O level of AND gate

V_A / V	V_B / V	V_F / V	Logic value of F
0.0	0.0	0.4	F
0.0	4.7	0.4	F
4.7	0.0	0.4	F
4.7	4.7	4.7	T

Analysis: From the table we know that only when both A and B are in high level, F is in high level. In the logic aspect of view, F is true when both A and B are true, and it is false otherwise. This relation satisfies the formula $F = AB$.

5.2 Assemble an OR gate with diodes

The red numerical reading shows V_A while the blue one shows V_F . Since V_A and V_B are symmetric, we can assume that the high and low level of V_B is the same as V_A .

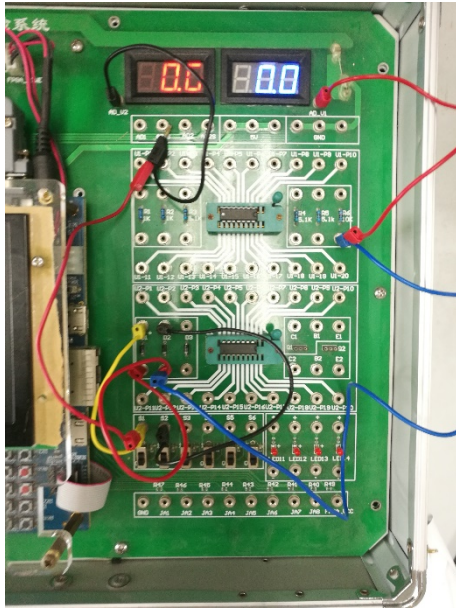


Figure 24 OR gate (A: low B: low)

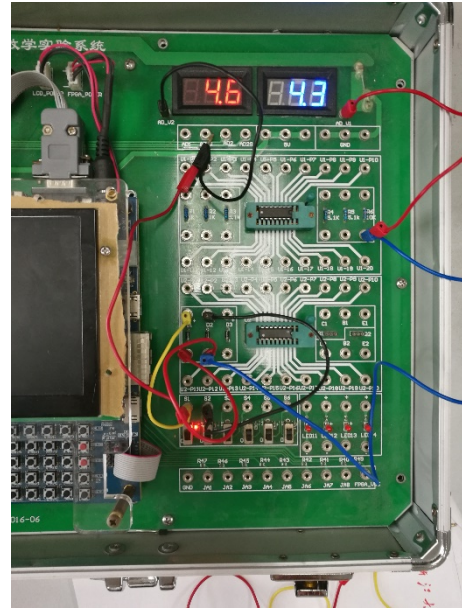


Figure 25 OR gate (A: high B: low)

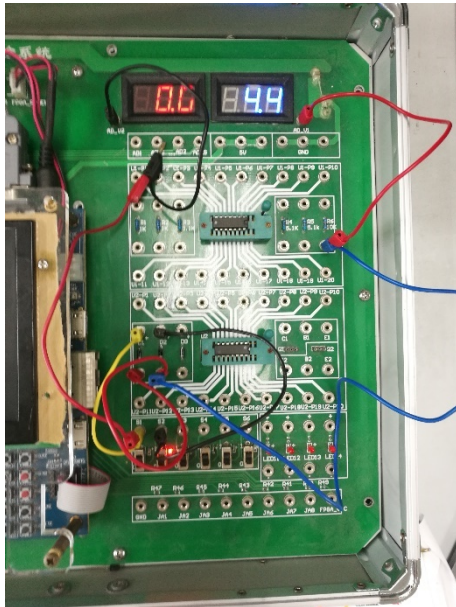


Figure 26 OR gate(A: low B: high)

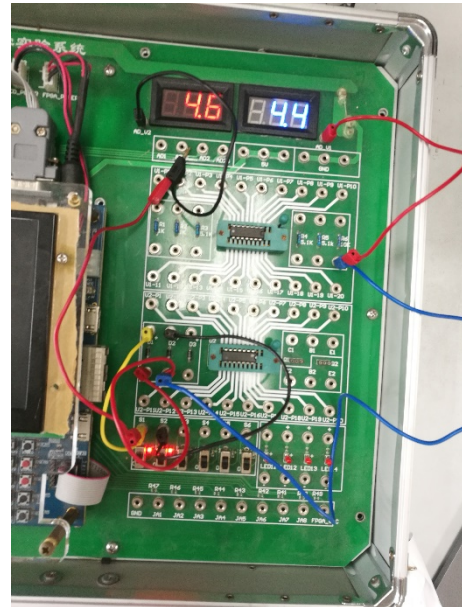


Figure 27 OR gate(A: high B: high)

Figure 28 I/O level of OR gate

V_A / V	V_B / V	V_F / V	Logic value of F
0.0	0.0	0.0	F
4.6	0.0	4.3	T
0.0	4.6	4.4	T
4.6	4.6	4.4	T

Analysis: From the table we know that only when both A and B are in low level, F is in low level. In the logic aspect of view, the value of F is false when both A and B are false, and it is true otherwise. This relation satisfies $F = A + B$.

5.3 Assemble a NOT gate with a Triode.

The red numerical reading shows V_A while the blue one shows V_F .

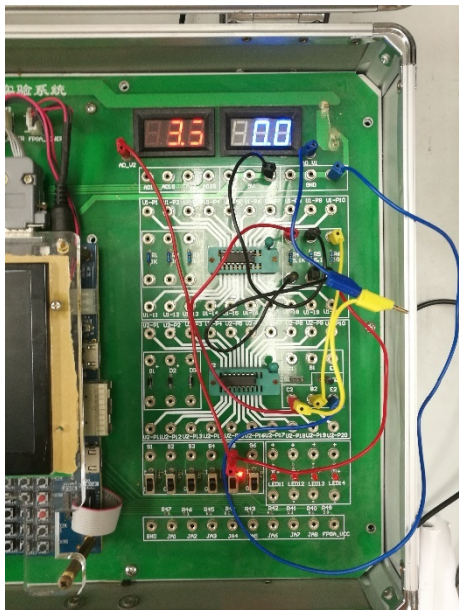


Figure 29 NOT gate (A: high)

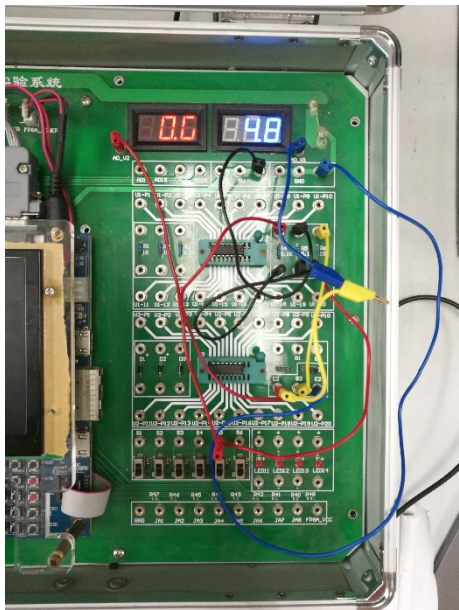


Figure 30 NOT gate (A: low)

Figure 31 I/O level of NOT gate

V_A / V	V_F / V	Logic value of F
3.3	0	F
0	4.8	T

Analysis: When A is in high level, F is in low level. Conversely, when A is in low level, F is in high level. In logic aspect of view, F has opposite logic value to A. This relation satisfies $F = \overline{A}$.

5.4 Assemble a NAND gate with transistors.

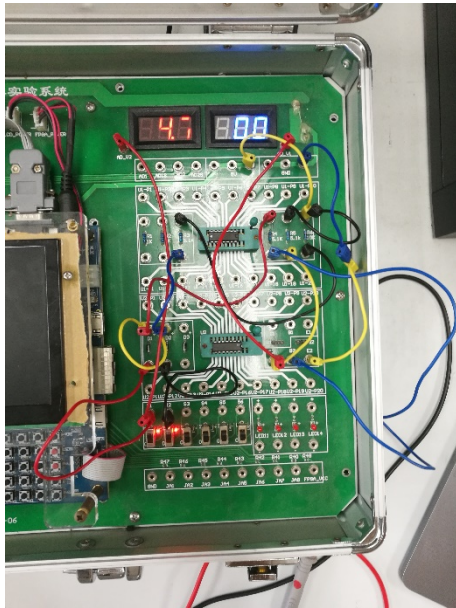


Figure 32 NAND gate (A: high B: high)

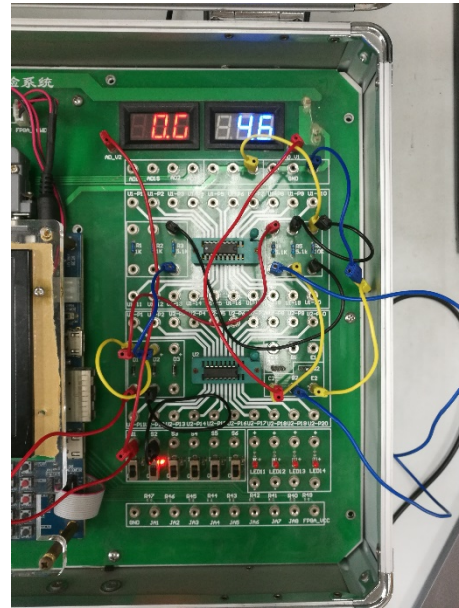


Figure 33 NAND gate (A: low B: high)

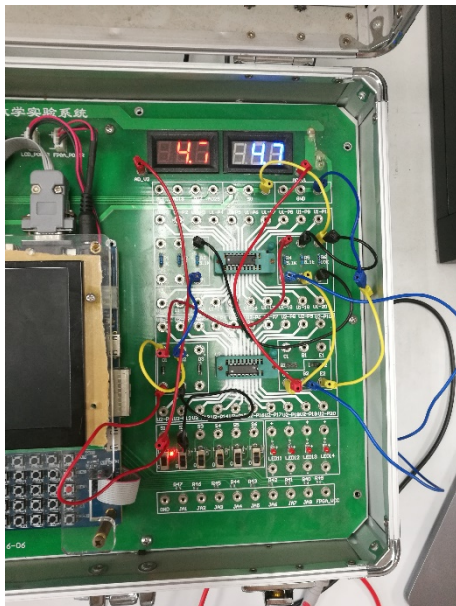


Figure 34 NAND gate (A: high B: low)

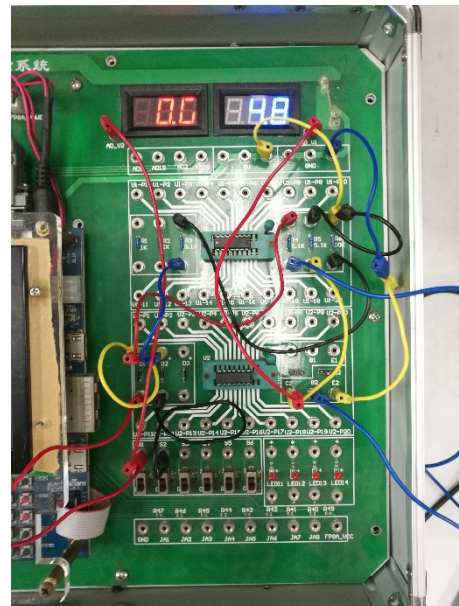


Figure 35 NAND gate (A: low B: low)

Figure 36 I/O level of NAND gate

V_A / V	V_B / V	V_F / V	Logic value of F
4.7	4.7	0	F
0.0	4.7	4.6	T
4.7	0.0	4.7	T
0.0	0.0	4.8	T

Analysis: When both A and B are in high level, F is in low level. Otherwise F is in high

level. In the logic aspect of view, when both A and B are true, F is false, otherwise F is true. This relation satisfies $F = \overline{AB}$.

6. Discussion and Revision of the Experiment

Last semester we learnt about logic operations in discrete mathematics. But at that time, it wasn't clear to me how to apply these in circuits and moreover, in computers. This experiment enlightened me how to turn logic operands to electric levels and logic operations to gate circuits. In this it's very ingenious and impressing to use the semiconducting characteristics of diodes and triodes to design these logic gates.

The experiment itself is rather simple, and I finished most of it with my partner without consuming much time. The only problem we met was that when we finished the NAND gate, it didn't work. We checked for some time and in the end, we found out that one of the wires was cut off. The circuit itself was correctly connected. I think the kernel of this experiment is to understand how the semiconductors work and how the output of the gates changes as the input changes. Once you fully comprehend the principles of the gates, you just assemble the circuits according to the diagrams and everything will be uncomplicated.

Lab 3 Functions of Integrated Logic Gate Circuit and Parameter Test

1. Purposes and Requirements of the Experiment

- 1.1 To get familiar with the functions, external electrical characteristics and special usages of basic logic gate circuits.
- 1.2 To get familiar with packaging and pin functions of TTL NAND gate and MOS NOR gates.
- 1.3 To master how to measure the major parameters and static characteristics, and to enhance our comprehension of the parameters' significance.
- 1.4 To intensify the concept that signal propagation has a time delay.
- 1.5 To get more familiar with instruments like oscilloscope, function signal generator, etc.

2. Tasks and Principles of the Experiment

2.1 Tasks

1. Verify the logic functions of integrated circuit 74LS00 NAND gate.
2. Verify the logic functions of integrated circuit CD4001 NOR gate.
3. Measure the propagation delay t_{pd} of integrated circuit 74LS00 NAND gate.
4. Measure the propagation delay t_{pd} of integrated circuit CD4001 NOR gate.
5. Measure the propagation characteristics of integrated circuit 74LS00 and its on-level V_{on} and off-level V_{off} .

2.2 Principles

1. Propagation time is a dynamic parameter. It describes the time delay between the input signal and its corresponding output signal, and it's caused by the charging and discharging time of transistors' PN-node capacitance, distributed parasitic capacitance, load capacitance and other capacitances.

Average propagation time consists of two parts: one is the time delay of the transition from high level to low level t_{PHL} , and the other is the time delay of the transition from low level to high level t_{PLH} .

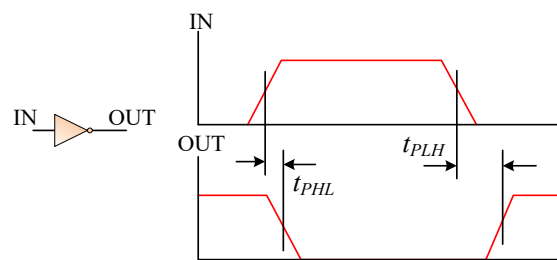


Figure 37 Two parts of propagation time

Usually we take the 50% medium points of the maximum and minimum voltages as the reference points of average propagation delay time. We measure t_{PHL} and t_{PLH} and then take their average as the average propagation delay time.

To enhance the accuracy, we use ring oscillator to measure propagation delay time. Assume that every NAND gate has the same delay time, then the oscillating period $T = 6 t_{pd}$. The delay time is $T / 6$.

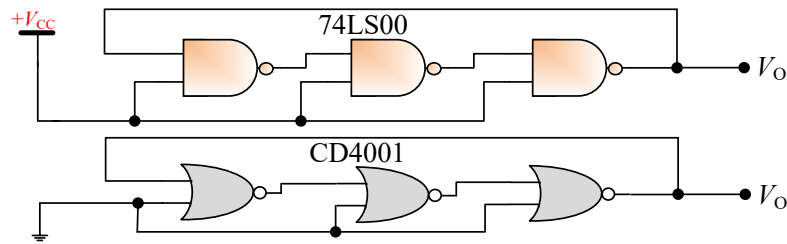


Figure 38 Ring oscillators of NAND and NOR gates

2. Voltage-transfer characteristic is the relationship of the output voltage to change as the input voltage changes. The off-level V_{OFF} describes the highest low input level when the output level just reaches the rated level of the other state. While the on-level V_{ON} describes the lowest high input level when the output level just reaches the turning point of jumping to the rated level of the other state.

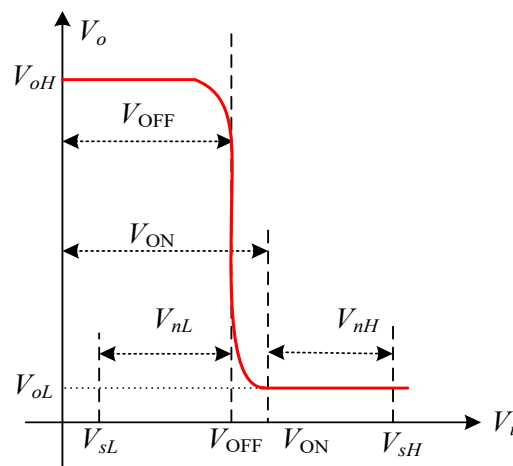


Figure 39 Voltage-transfer characteristics of gate 74LS00

3. Mainly-used Instruments and Materials

Instruments

1	digital oscilloscope RIGOL-DS162	1
2	digital multimeter	1
3	experiment box of logic circuit designing	1

Materials

4	two-input NAND gate 74LS00	1
5	two-input NOR gate CD4001	1
6	resistances	

6.1 4.7k Ω potentiometer 1

6.2 100 Ω / 1kW 1

4. Procedures and Operations of the Experiment.

4.1 Verify the logic functions of NAND gate 74LS00

1. Plug the chip into the IC socket in the experiment box. NOTICE the direction of the chip.
2. Connect the circuits according to the diagram. Connect V_{CC} To 5V and connect GND to ground electrode.
3. The high and low level are generated by switches S14-S17.
4. Traverse to input all combinations of A and B. Measure the voltage V_A , V_B and the output voltage V_F . Record them into the table.

Figure 40 I/O level of 74LS00

V_A (V)	V_B (V)	V_F (V)

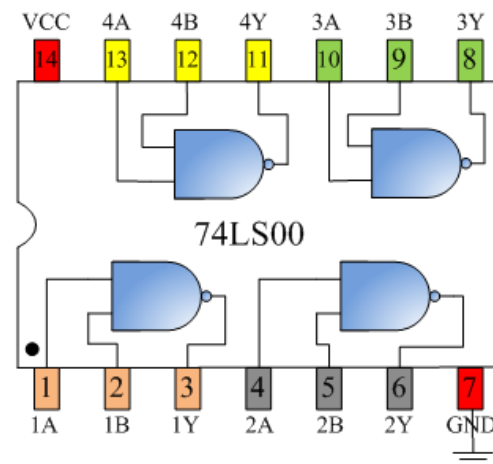


Figure 41 Inside structure of 74LS00

4.2 Verify the logic functions of NOR gate CD4001

1. Plug the chip into the IC socket in the experiment box.
2. Connect the circuits according to the diagram. Connect V_{CC} To 5V and connect GND to ground electrode.
3. The high and low level are generated by switches S14-S17.
4. Traverse to input all combinations of A and B. Measure the voltage V_A , V_B and the output voltage V_F . Record them into the table.

- Repeat step 3-4, measure the logic relationship of the other 3 gates and justify whether they function well.

Figure 42 I/O level of CD4001

V_A (V)	V_B (V)	V_F (V)

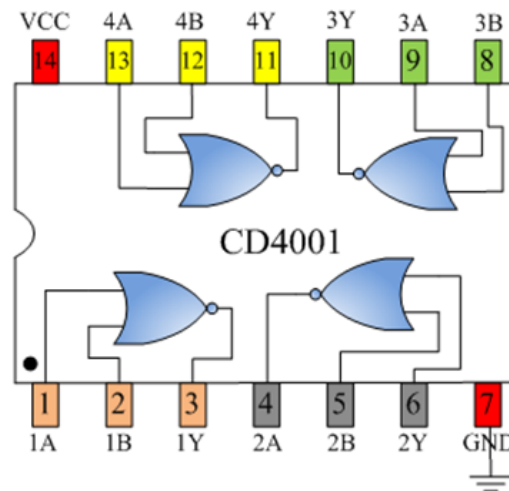


Figure 43 Inside structure of CD4001

4.3 Measure the propagation delay of logic gate 74LS00

- Plug the chip into the IC socket in the experiment box. NOTICE the direction of the chip.
- Connect the circuits according to the diagram. Connect V_{CC} To 5V and connect GND to ground electrode.
- Connect the oscilloscope to any of the input ends and output ends of the oscillator.
- Switch the frequency knob and measure the wave shape of V_o . Read the period T and calculate the propagation delay time (normally between 30-60ns).

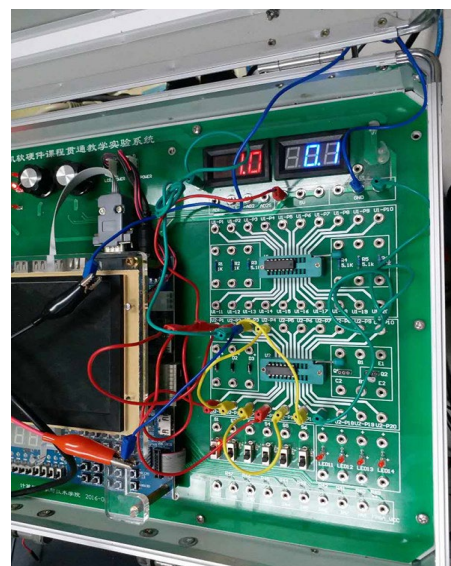
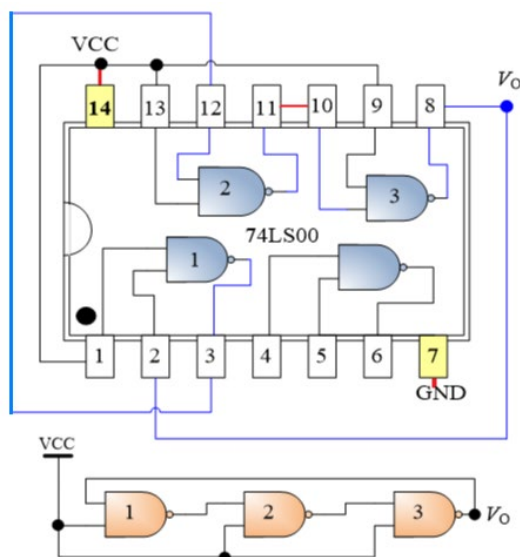


Figure 44 Measure the propagation delay of 74LS00

4.4 Measure the propagation delay of logic gate CD4001

1. Plug the chip into the IC socket in the experiment box. NOTICE the direction of the chip.
2. Connect the circuits according to the diagram. Connect V_{CC} To 5V and connect GND to ground electrode.
3. Connect the oscilloscope to any of the input ends and output ends of the oscillator.
4. Switch the frequency knob and measure the wave shape of V_o . Read the period T and calculate the propagation delay time (normally between 500-1000ns).

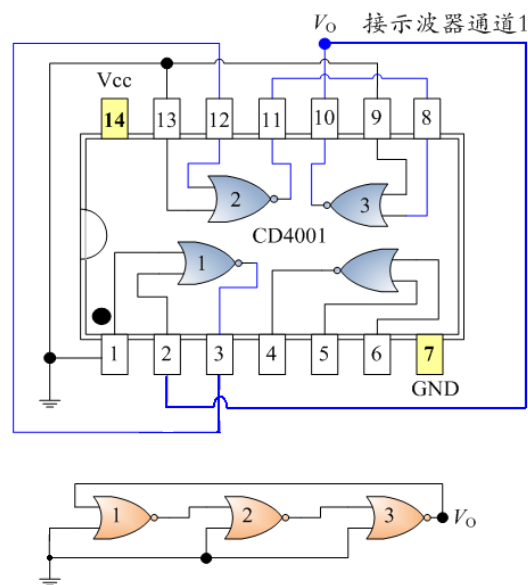


Figure 45 Measure the propagation delay of gate CD4001

4.5 Measure the propagation characteristics and on/off-level of 74LS00

1. Plug the chip into the IC socket in the experiment box.
2. Connect the circuit according to the diagram.
3. Connect the DC electric meters to A and the output of the NOR gate 2Y separately.
4. Slowly adjust potentiometer W from end a to end b. Observe the readings of V_i and V_o , and record them into the table.
5. Plot a diagram according to the data in the table, and calculate V_{ON} and V_{OFF} .

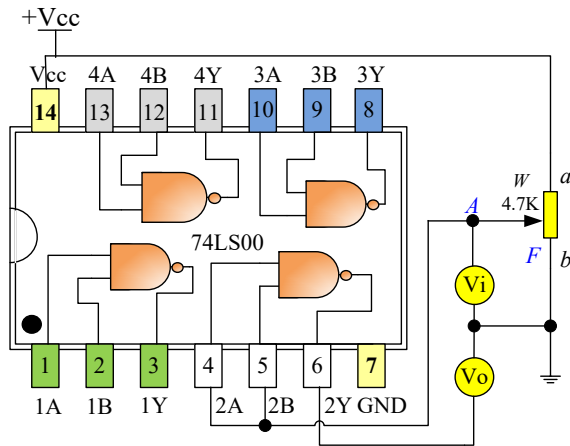


Figure 46 Diagram of measuring 74LS00

V_i/V	V_o/V	V_i/V	V_o/V
0		:	:
0.2		:	:
0.4		:	:
0.6		:	:
0.8		2.0	
:	:	2.5	
:	V_{OFF}	3.0	
:	:	3.5	
:	:	4.0	
:	V_{ON}	4.5	
:	:	5.0	

Figure 47 Propagation characteristic of 74LS00

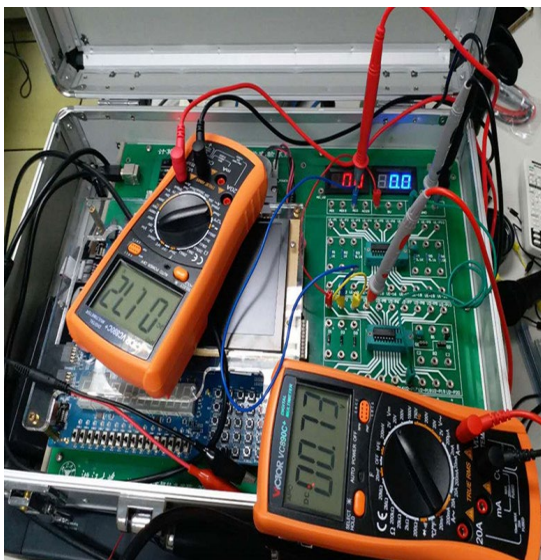
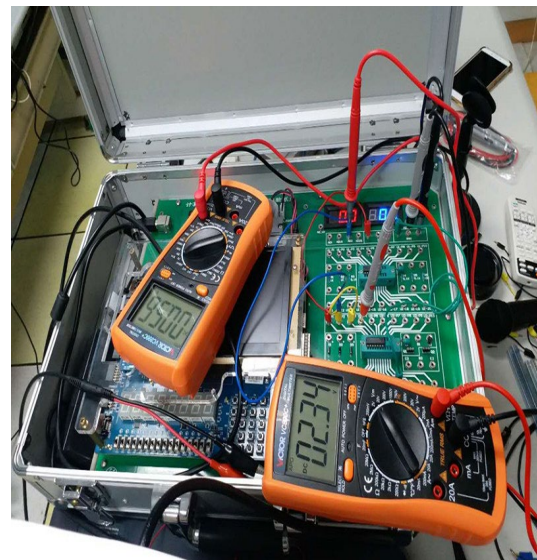


Figure 48 Measuring propagation characteristic of 74LS00



5. Analysis and Result of the Experiment

5.1 Verify the logic functions of gate 74LS00

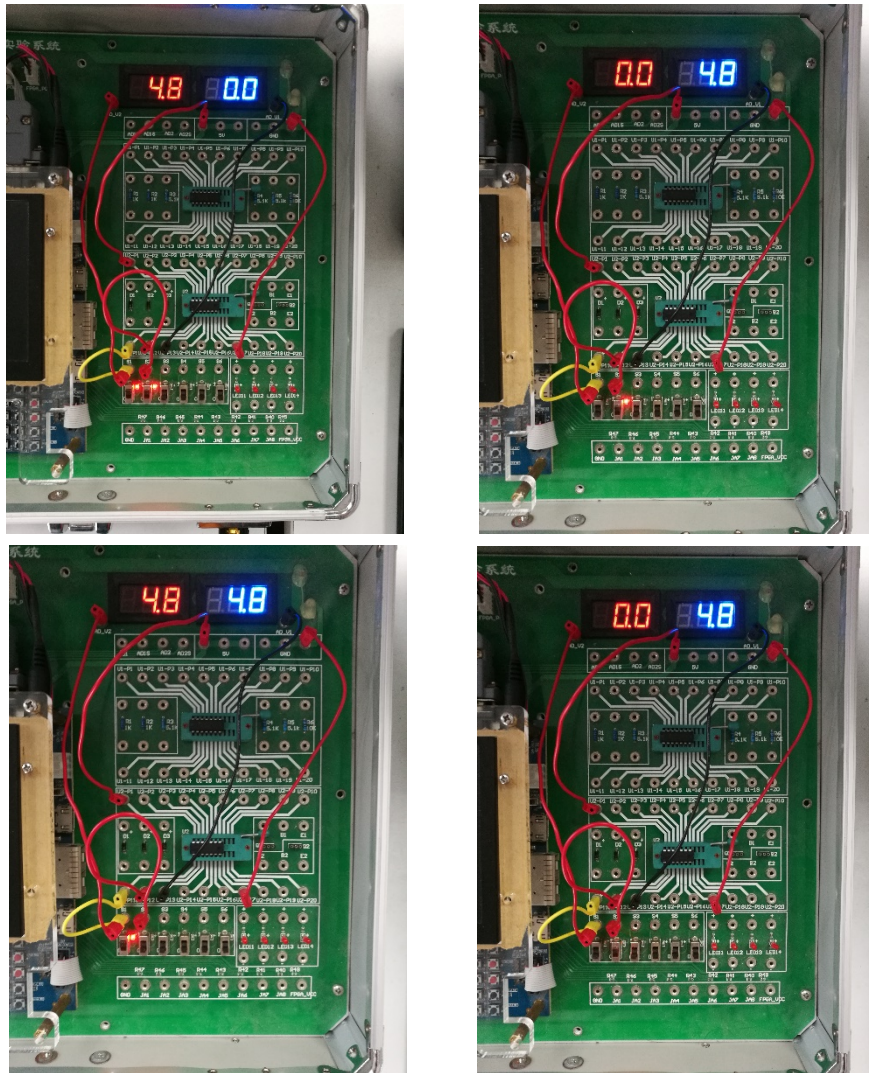


Figure 49 Four states of 74LS00

Figure 50 I/O level of 74LS00 NAND gate

V_A (V)	V_B (V)	V_F (V)
0	0	4.8
0	4.8	4.8
4.8	0	4.8
4.8	4.8	0

Analysis: The I/O levels of 74LS00 NAND gate satisfy $F = \overline{AB}$. It functions well as a NAND gate.

5.2 Verify the logic functions of gate CD4001

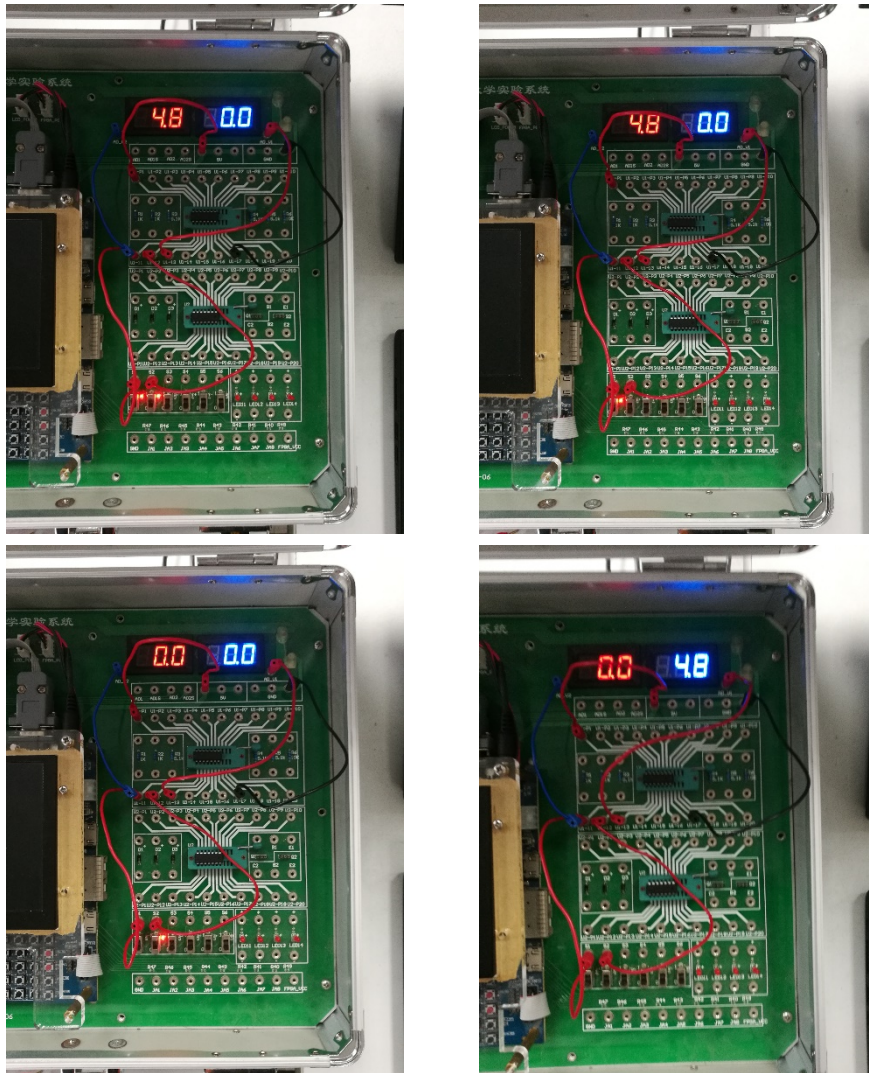


Figure 51 Four states of CD4001

Figure 52 I/O level of CD4001

V_A (V)	V_B (V)	V_F (V)
0	0	4.8
0	4.8	0
4.8	0	0
4.8	4.8	0

Analysis: The I/O levels of CD4001 NOR gate satisfy $F = \overline{A + B}$. This indicates that it functions well as a NOR gate.

5.3 Measure the propagation delay of logic gate 74LS00

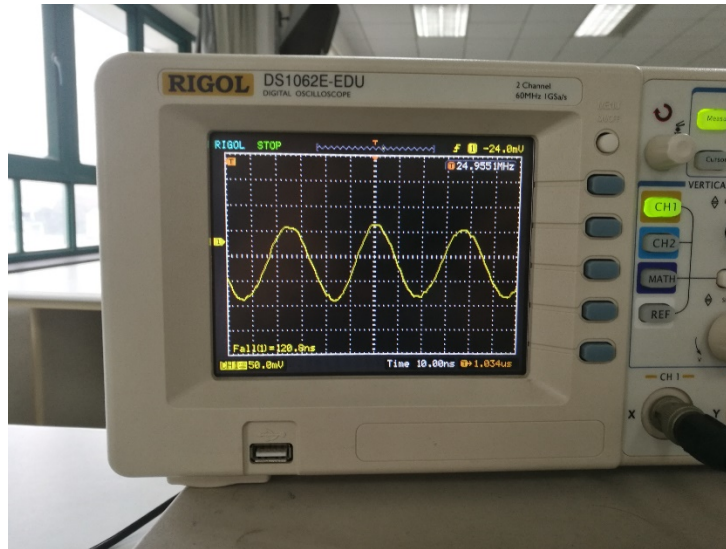


Figure 53 Wave of the oscillation of gate 74LS00

Analysis: The parameters measured and propagation time computed are listed in the following table.

Figure 54 Measuring propagation time of 74LS00

Oscillator reading	Sensitivity	Computed Period	t_{pd}
3.7 Div	10ns/Div	37ns	6.17ns

5.4 Measure the propagation delay of logic gate CD4001

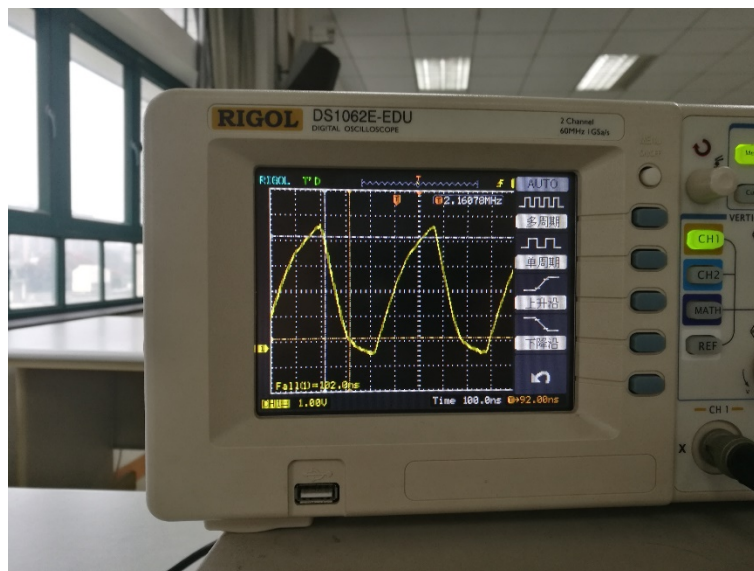


Figure 55 Wave of the oscillation of gate CD4001

Analysis: The parameters measured and propagation time computed are listed in the following table.

Figure 56 Measuring propagation time of CD4001

Oscillator reading	Sensitivity	Computed Period	t_{pd}
4.8 Div	100ns/Div	480ns	80ns

5.5 Measure the propagation characteristics and on/off level of 74LS00

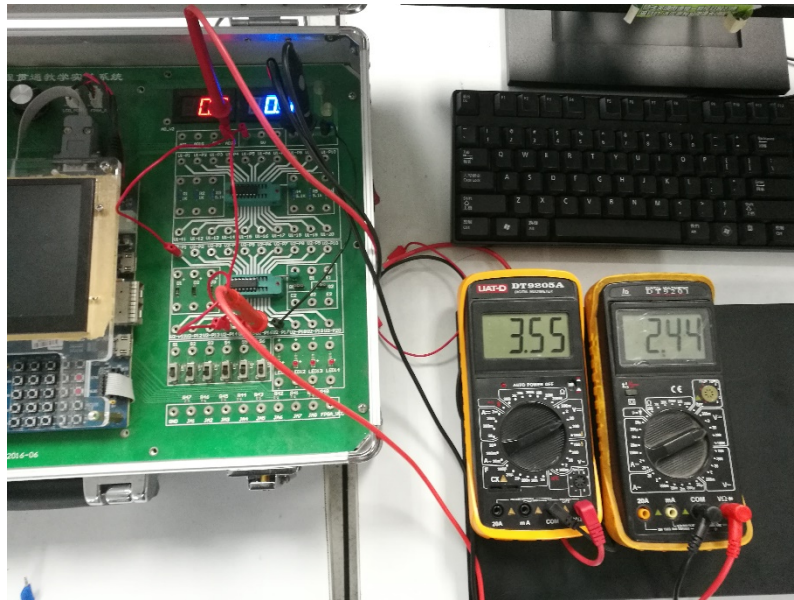


Figure 57 Measuring the propagation characteristic of 74LS00

Figure 58 Propagation characteristic of 74LS00

V_i/V	V_o/V	V_i/V	V_o/V
0.04	4.93	2.6	3.03
0.2	4.93	2.8	2.87
0.4	4.93	3.0	2.70
0.6	4.93	3.2	2.29
0.8	4.93	3.4	0
1.0	4.93	3.6	0
1.2	4.93	3.8	0
1.4	4.93	4.0	0
1.6	4.93	4.2	0
1.8	4.93	4.4	0
2.0	4.93	4.6	0
2.2	4.93	4.8	0
2.4	4.93	4.87	0

Analysis: We plot the propagation characteristics data in the following diagram.

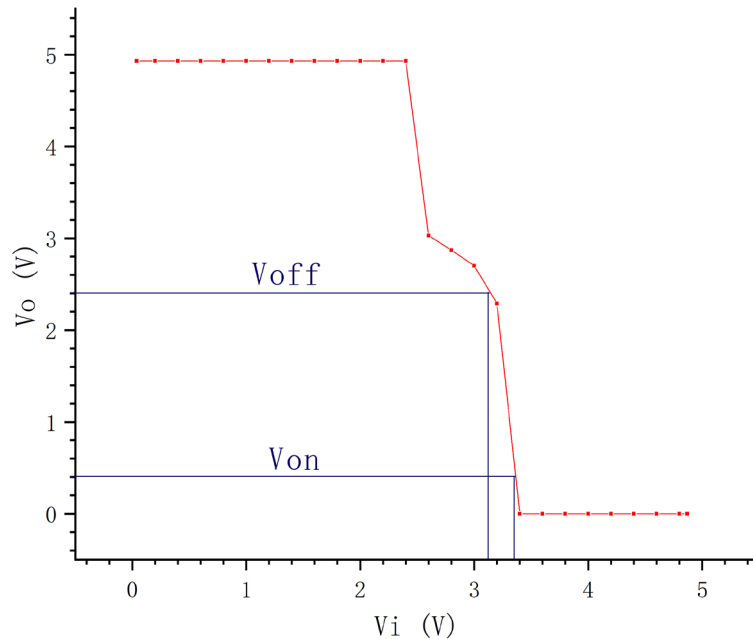


Figure 59 Diagram of propagation characteristics of 74LS00

From this graph we can roughly get V_{off} and V_{on} .

$$V_{\text{off}} = 3.14\text{V}$$

$$V_{\text{on}} = 3.36\text{V}$$

6. Discussion and Revision of the Experiment.

This experiment is an extension of the last one. It illustrates concepts like propagation delay time and on/off voltages and by doing so, it deepens our comprehension of logic gates circuits.

Meanwhile, this is the first time we measure something meaningful with the oscilloscope. When we tried to measure the propagation time of 74LS00 for the first time, the curve the oscilloscope displayed was not stable. It leaped rapidly and violently and we weren't able to measure it. We checked the circuits and found that two of the wires were not connected tightly. After repairing it the curve was much more stable and we pressed the STOP button to measure the period. The period we got was close to the standard value and we assumed it was correct. But still we have a doubt: we found that the amplitude of the wave is changing periodically. It seemed that the wave is somehow a beat which is superposed by two waves and we didn't know why. We referred to some documents and inferred that this was caused by the attributes of the oscillator, but we didn't make it to find a confirmed answer.

Lab 4 Application of EDA experiment platform and environment

1. Purposes and Requirements of the Experiment

1. To get familiar with Verilog HDL language and to use it to build basic logic elements.
To use Xilinx ISE platform to conduct inputting, editing, debugging, behavior simulation and synthesized post-functional simulation.
2. To get familiar with SWORD FPGA developing platform, and conduct sequential constraints, pin constraints and sequential simulation after mapping the wiring on ISE platform.
3. To use Xilinx ISE to download the verified code onto the experiment board and verify it on the board.

2. Tasks and Principles of the Experiment

1. Get familiar with the operating environment and installing process of ISE tool software.
2. Design simple combination logic circuits. Use graphical input to input descriptions of logic functions. Build an FPGA to realize the Xilinx ISE designing management project of digital system, and conduct editing, debugging, compiling, behavior simulation, sequential constraints, pin constraints, sequential simulation after mapping the wiring and downloading and verifying the FPGA programming code.
3. Design simple sequential logic circuits. Use Verilog code input to input descriptions of logic functions. Build an FPGA to realize ISE designing management project of digital system, and conduct editing, debugging, compiling, behavior simulation, sequential constraints, pin constraints, sequential simulation after mapping the wiring and downloading and verifying the FPGA programming code.

3. Mainly-used Instruments and Materials

Instruments

- | | |
|-------------------------------------|---|
| 1. Computer installed with ISE 14.7 | 1 |
| 2. SWORD developing board | 1 |

4. Introduction to ISE 14.7 platform with Verilog HDL

In this section we use a specific example to illustrate how to use Verilog language to describe logical functions of a circuit on ISE 14.7 platform.

4.1 Problem and solution

Problem: The passageways of a three-story building share a single lamp. Each floor is furnished with a button switch that can control the lamp independently. Every time the button is pressed, the lamp stays on for several seconds and turns off automatically. Please design the controlling circuit of the lamp.

First, we assign values to the variables. Let S_1, S_2, S_3 be the input of the button switches and F be the output of the circuit. The input value is 1 when the switch is closed and 0 when it's open. The output value is 1 if the lamp is on and 0 if it's off.

We can write Verilog code as follows.

```
module lamp_control(input wire clk,
    input wire S1,
    input wire S2,
    input wire S3,
    output wire F
);

    parameter C_NUM = 28;
    parameter C_MAX = 28'hFFF_FFFF;

    reg [C_NUM-1:0] count;
    wire [C_NUM-1:0] c_next;

    initial begin //initialization
        count = C_MAX;
    end

    assign w=S1^S2^S3;

    assign F = ((count < C_MAX) ? 1'b1 : 1'b0);

    always@(posedge clk)
    begin
        if(w == 1'b1)
            count = 0;
        else if(count < C_MAX)
            count = c_next;
    end

    assign c_next = count + 1'b1;
endmodule
```

Now we implement it in ISE 14.7.

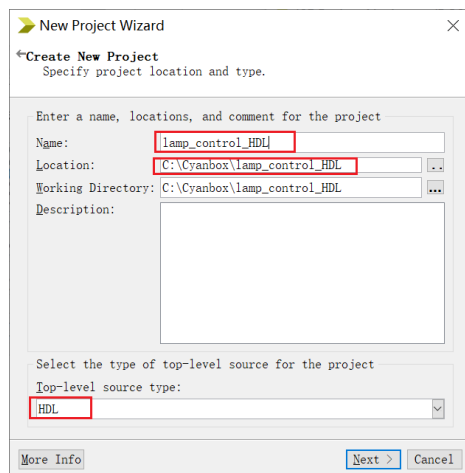
4.2 Implementation on ISE 14.7

1. Build a lamp-controlling project lamp_control_HDL.ise

- Open ISE Design Suite 14.7, click File -> New Project
- Modify the settings as follows in the dialog box:

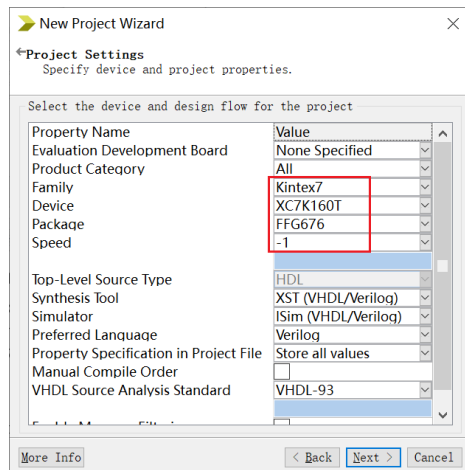
Project Name: lamp_control_HDL
Top-Level Source Type: HDL

Choose the path where the project is stored.



- Click Next to get the Project Settings page. Modify the settings as follows:

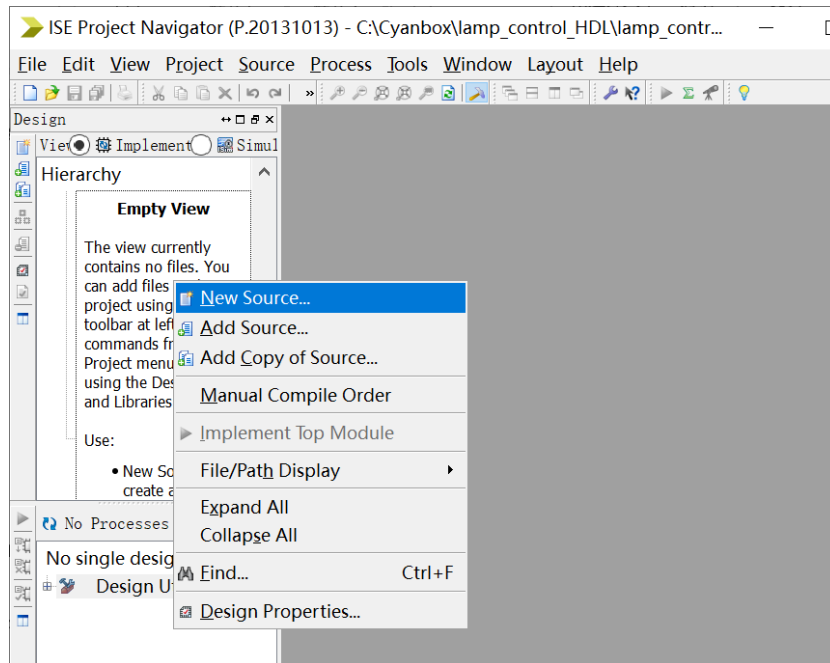
Family: Kintex7
Device: XC7K160T
Package: FFG676
Speed: -1



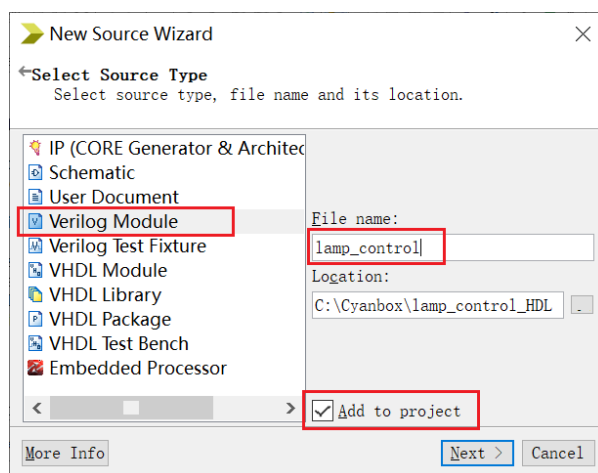
- Check it and click Next -> Finish to finish the building of the new project.

2. Create Verilog source file lamp_control.v

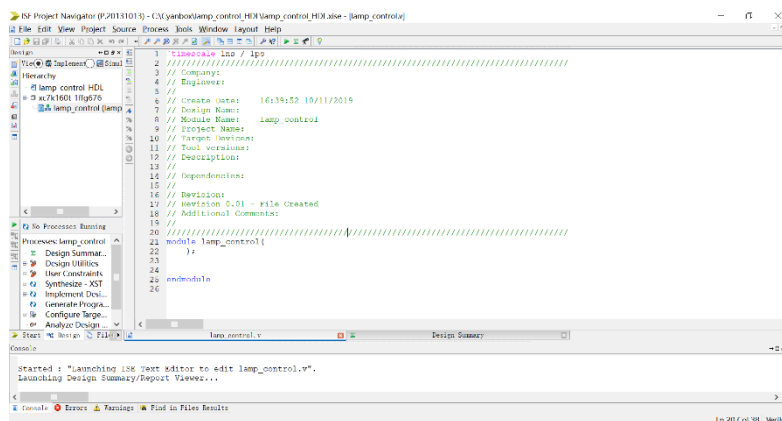
- Right click at the blank part of Design box, and click New Source in the menu.



- Select source type as Verilog Module. Type in the filename lamp_control. Tick on Add to project.

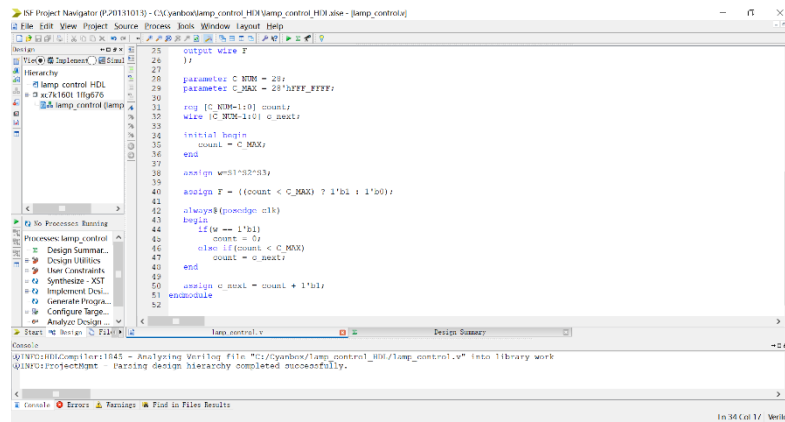


- Click Next continuously until finishing.



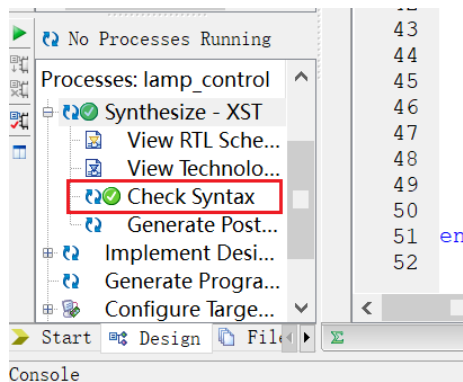
3. Edit the lamp-controlling Verilog HDL code

- Type in the code in the source code editor.



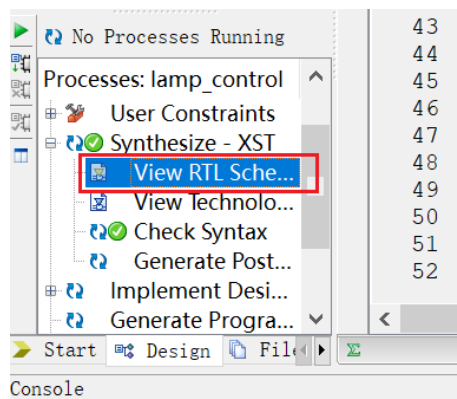
- Find Synthesize-XST -> Check Syntax in Processes box. Double click Check Syntax.

This is to check whether the codes conform to grammar rules of Verilog HDL. If there are any grammatical mistakes, correct them.



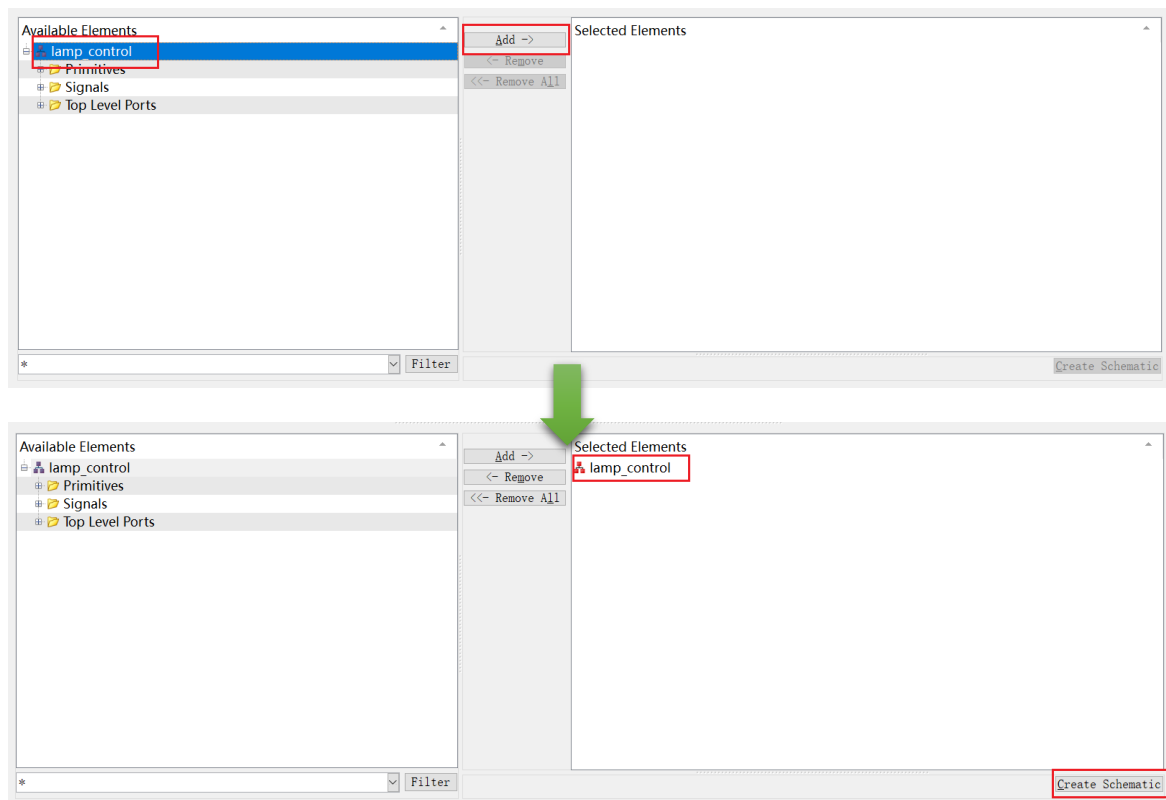
4. Synthesize the controlling code (optional)

- Select lamp_control.v in the Design box.
- Run Synthesis-XST -> View RTL Schematic in Processes box. Click OK in the Startup mode window.

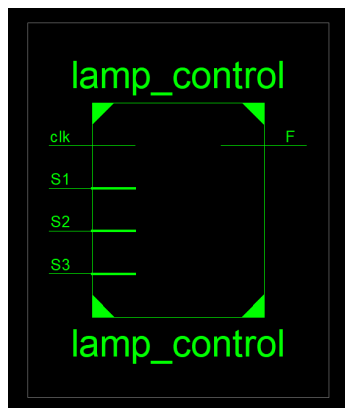


- Add lamp_control.v from Available Elements to Selected Elements. Then click

Create Schematic.



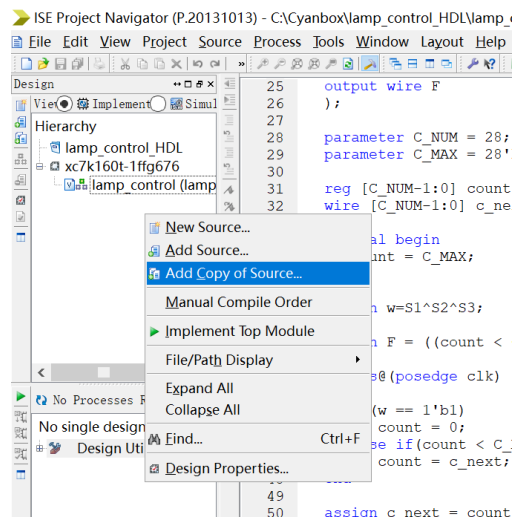
- Check whether the synthesized circuits matches the target of designing.



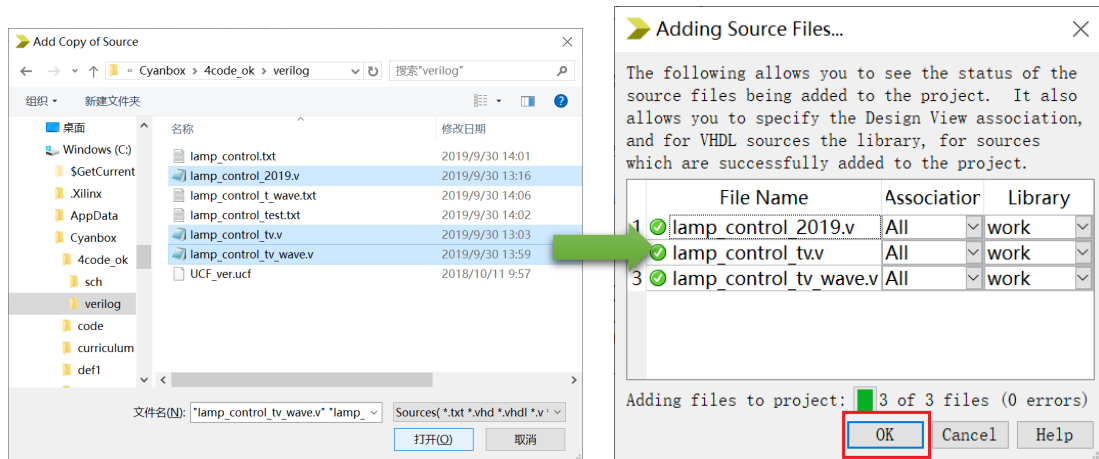
- Tips: Add copy of Source

During the process of the experiment, the time may be too tense for coding from cover to cover. You may add source files that have already been coded to the project.

- Right click at the blank of the Design box, choose Add Copy of Source.

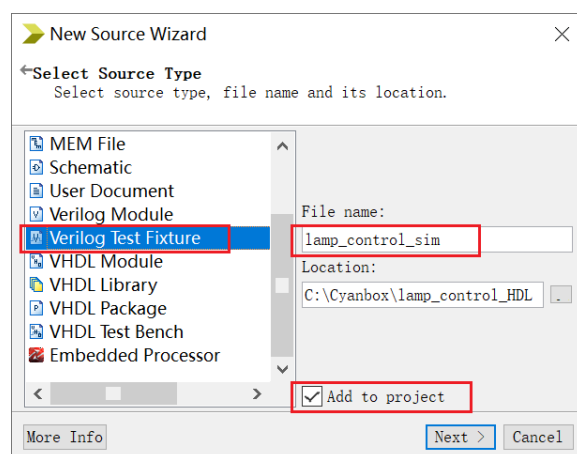


- Find the copy of source, and add them to the project



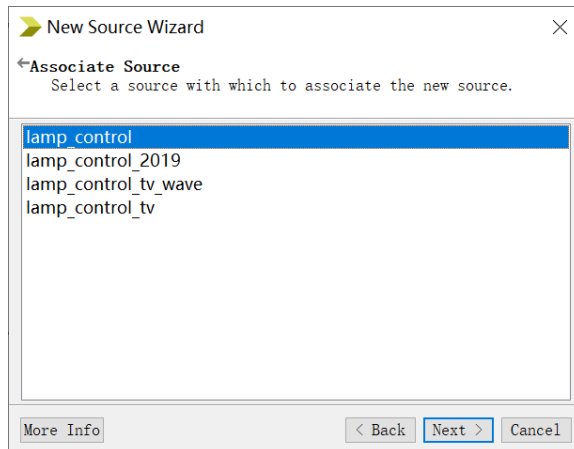
5. Build Benchmark Testing Waveform File.

- Right click at the blank of Source box and click New Source.
- Select source type as Verilog Test Fixture. Type in the filename lamp_control_sim and tick at Add to Project.



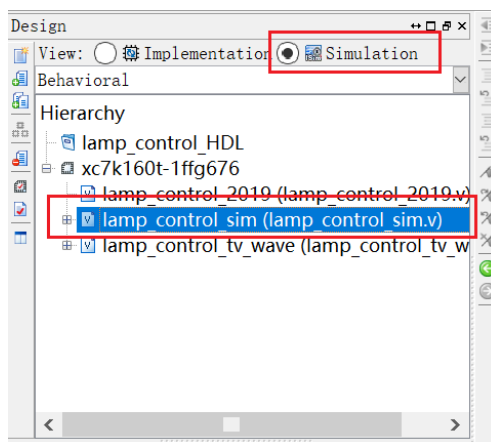
- If there are already several source code files, select the one you want the testing

file to associate with. Then click Finish to enter the editing window of lamp_control_sim.v.

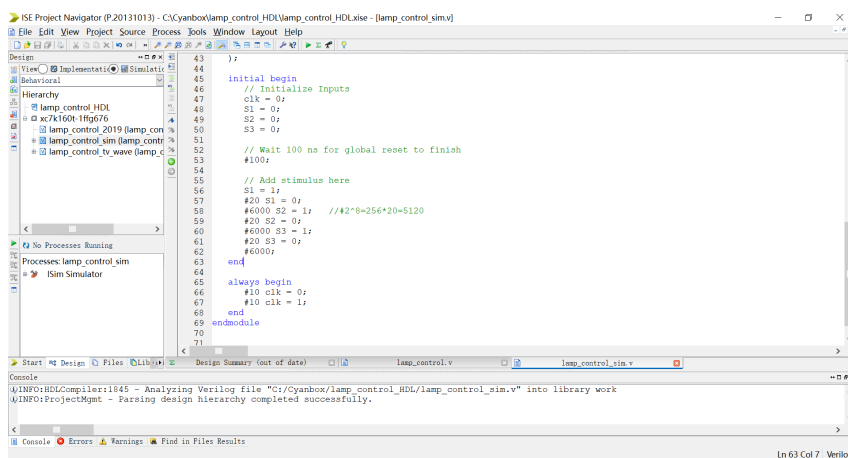


- Tips: How to find the sim testing file we just created?

lamp_control_sim.v is a testing waveform program. It can be found under Simulation View of the Design box.



- Edit the testing file.



- Change the bit number of the counter to 8 in lamp_control.v for ease of simulation.

```
parameter C_NUM = 28;
parameter C_MAX = 28'hFFF_FFFF;
```

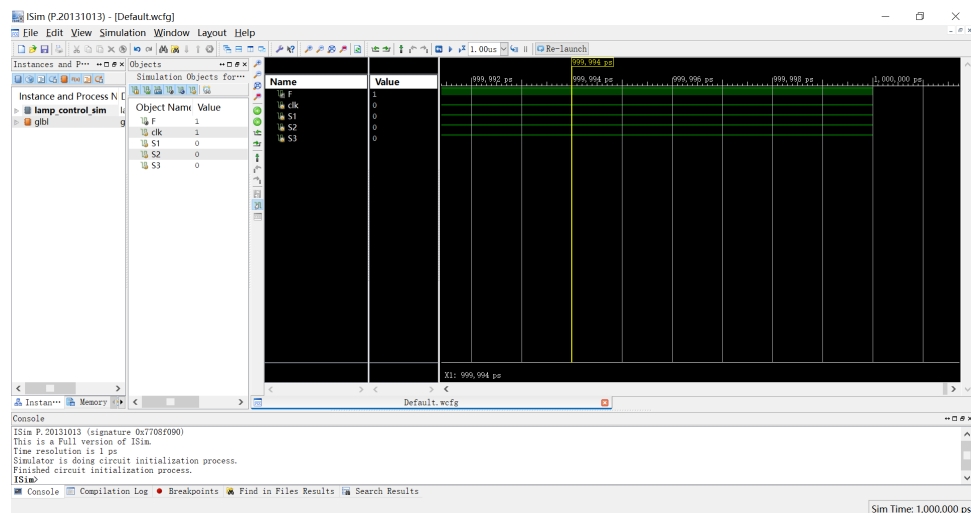
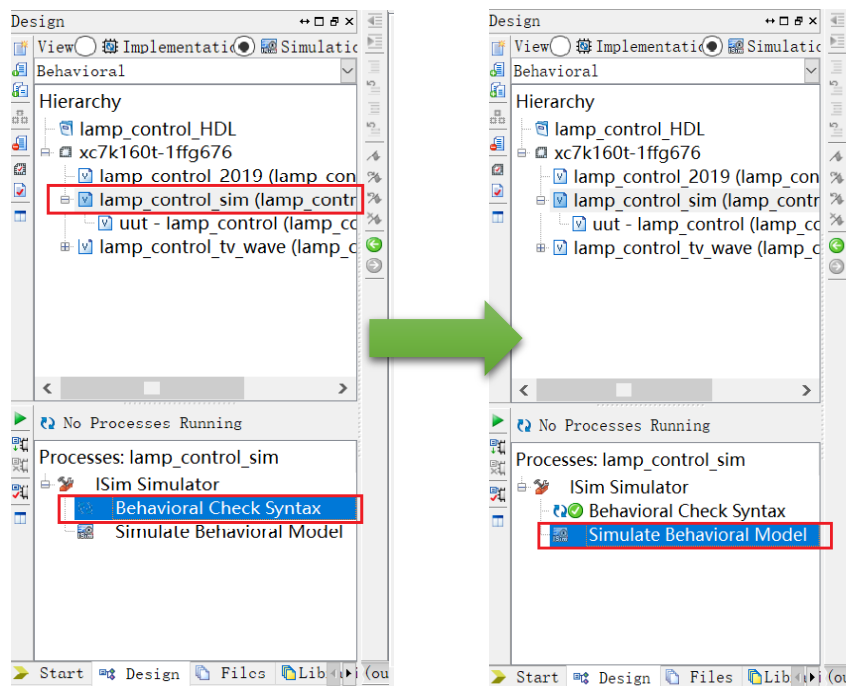
```
reg [C_NUM-1:0] count;
wire [C_NUM-1:0] c_next;
```

```
parameter C_NUM = 8;
parameter C_MAX = 8'hFF;
```

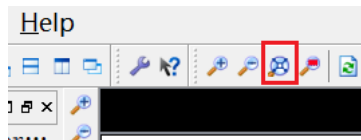
```
reg [C_NUM-1:0] count;
wire [C_NUM-1:0] c_next;
```

6. Run Simulation

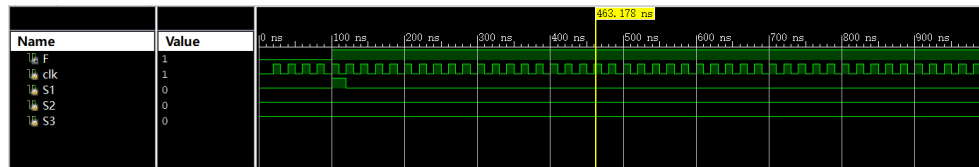
- Select lamp_control_sim.v in Design box, and double click Behavioral Check Syntax in Processes box. After passing the check, double click Simulate Behavioral Model to open simulation software ISim.



- The scale of time axis might be not appropriate to observe the waveform. We can press the Zoom to Full View tool in the toolbar to adjust the scale automatically.



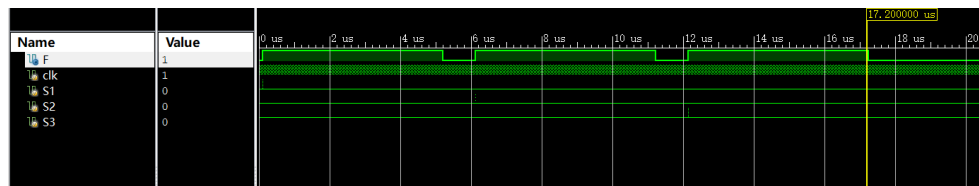
Now we can observe the waveform in 1 μ s.



We can also use Run for the time specified on the toolbar tool to control the running time of the simulated circuit. By using it together with Zoom to Full View and Restart we can observe how the waveform changes in a longer period of time. In practice, we start from 0 μ s, press Restart first, then Run for the time specified on the toolbar and Zoom to Full View.

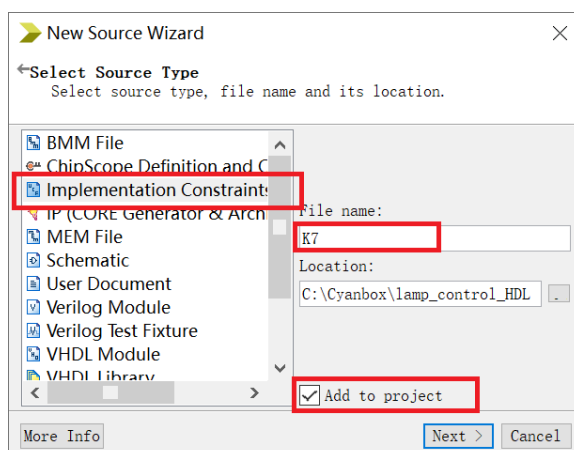


For example, we change the time period to 19 μ s and the waveform is as follows.



7. Build user sequential constraints and assign pin allocation to ports of the module

- Right click at the blank of Source box and click New Source in the menu.
- Select source type as Implementation Constraints File. Type in the filename K7 and tick at Add to Project.
- Click Finish to enter the editing window of K7.ucf.



- Type the following code into the file K7.ucf and save it.

```
NET"clk"LOC=AC18 | IOSTANDARD=LVCMOS18 ;
NET"S1"LOC=AA10 | IOSTANDARD=LVCMOS15;
NET"S2"LOC=AB10 | IOSTANDARD=LVCMOS15;
NET"S3"LOC=AA13 | IOSTANDARD=LVCMOS15;
NET"F"LOC=AF24 | IOSTANDARD=LVCMOS33 ;#D8
```

- Change the bit number of the counter from 8 to 28 in lamp_control.v.

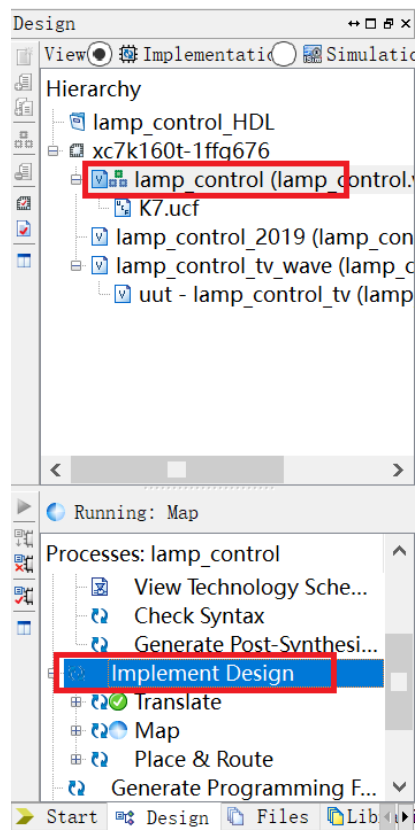
```
27
28 parameter C_NUM = 8;
29 parameter C_MAX = 8'hFF;
30
31 reg [C_NUM-1:0] count;
32 wire [C_NUM-1:0] c_next;
33
```



```
28 parameter C_NUM = 28;
29 parameter C_MAX = 28'hFFF_FFFF;
30
31 reg [C_NUM-1:0] count;
32 wire [C_NUM-1:0] c_next;
```

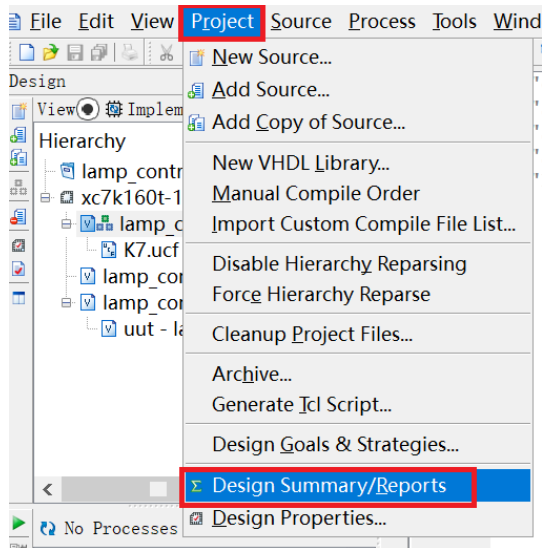
8. Implement design and download it onto the SWORD developing board.

- Select lamp_control in Design box and double click Implement Design in Processes box. Wait until the implementation process finishes.

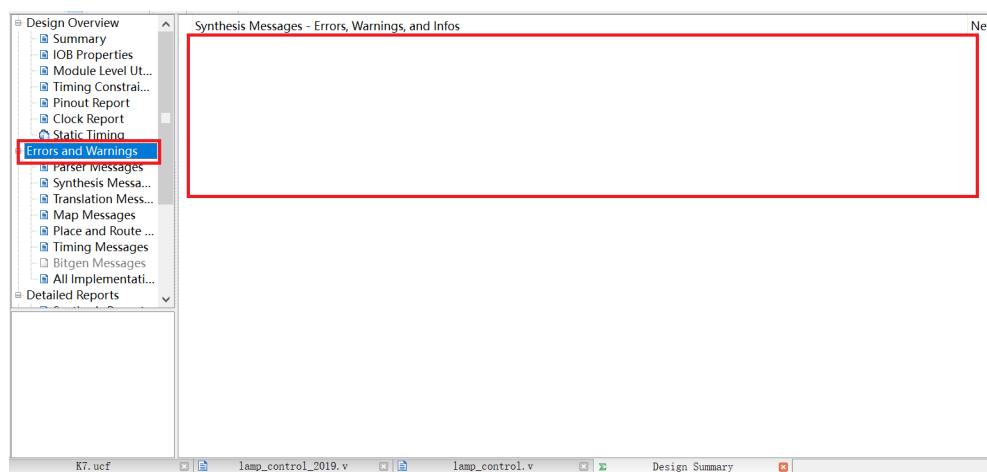


- After finishing the implementation, a Design Summary document will be generated.

You can find it in Project -> Design Summary/Reports.

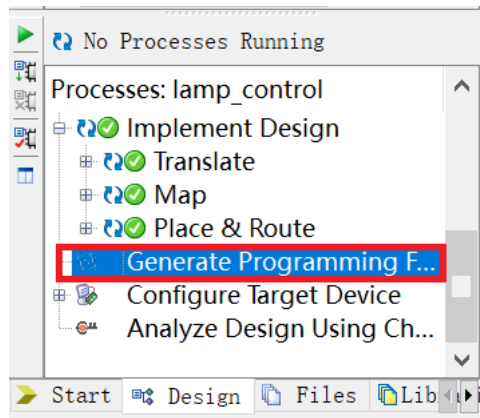


If there are any errors or warnings, you can find the reasons in the summary. Select Errors and Warnings in the catalogue on the left side of the window, and all errors and warnings generated during implementation will be listed with information about them.



Errors generated during synthesis originate from bugs in codes or diagrams. Errors and warnings generated during implementation originate from problems of ucf pin constraints.

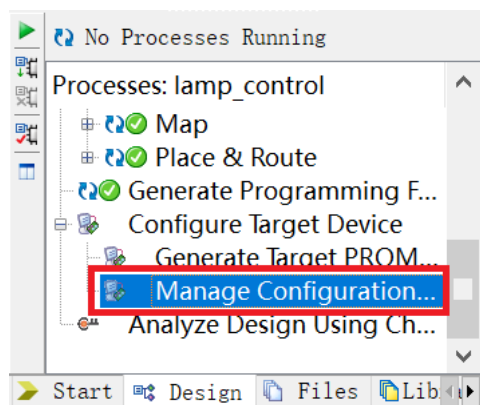
- When the design was successfully implemented, double click Generate Programming File to generate the Bit file which can be applied on the SWORD developing board.

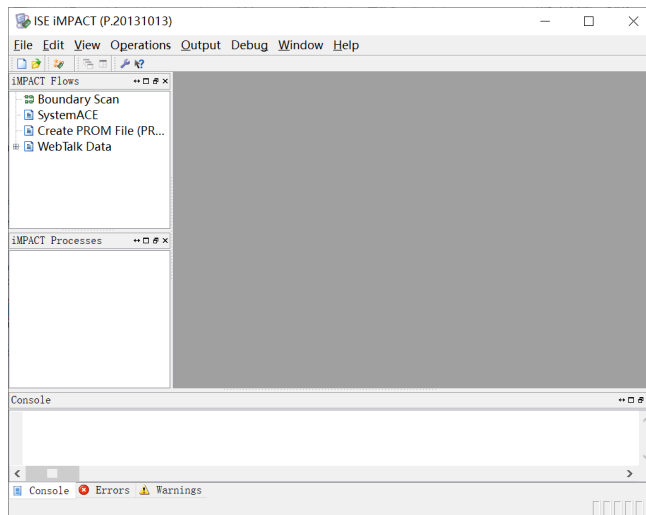


- When the generating process finishes, use the USB wire to connect the experiment box with your computer.

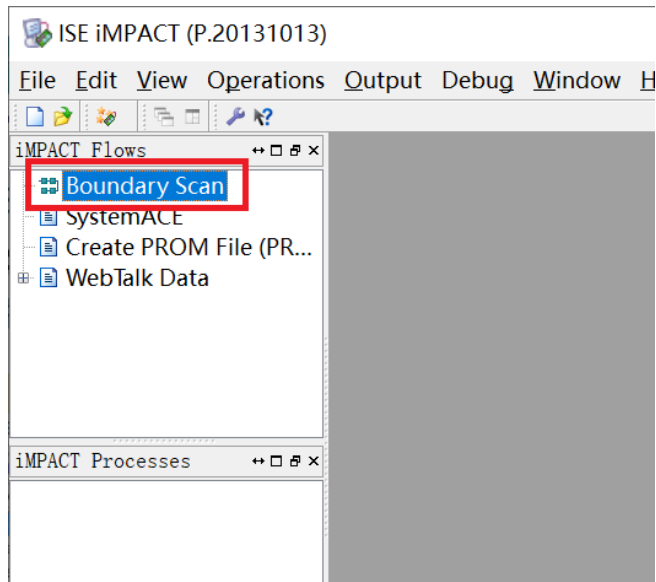


- Choose Implementation view in the Design box, and select lamp_control.v. Double click Manage Configuration Project (iMPACT), which is in Process -> Configure Target Device. The ISE iMPACT window will be launched.

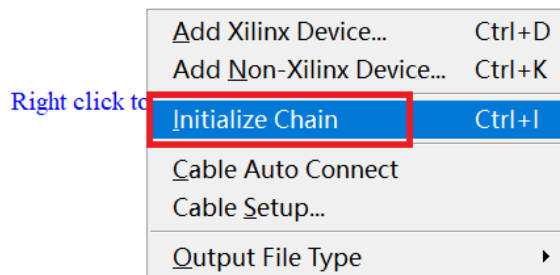




- Double click Boundary Scan. It will open the downloading and editing window.

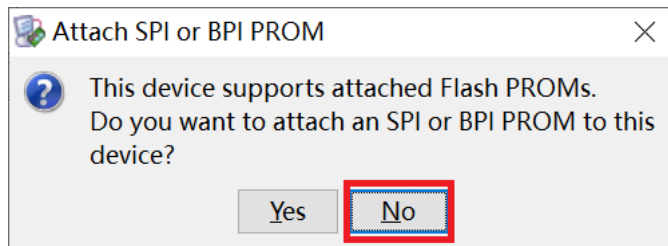
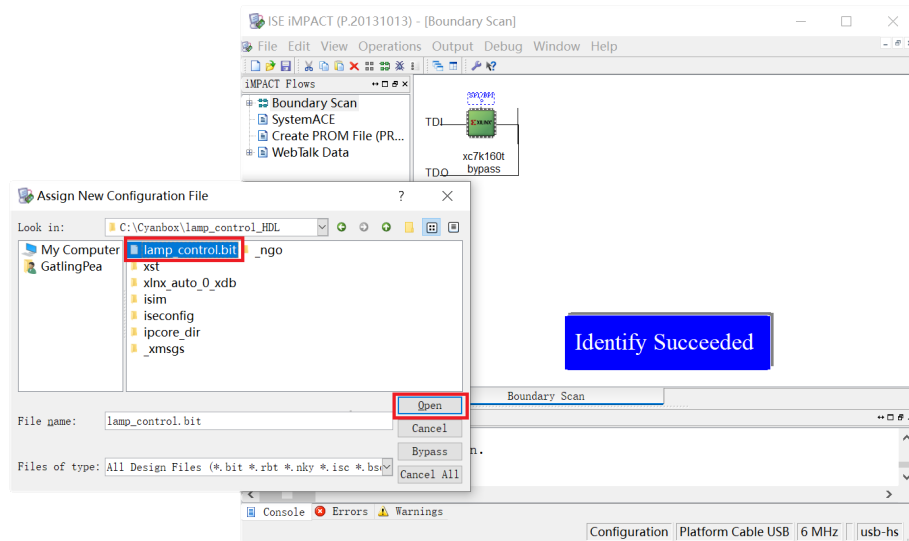


- Right click at the window, and click Initialize Chain. The system will automatically locate the JTAG downloading chain of the developing platform that has been attached to the computer. (Note that the SWORD board must be powered on now.)

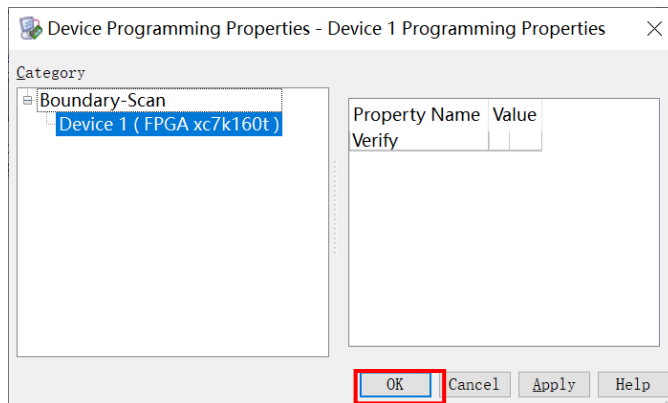


- When the initialization finishes, choose to continue and assign configuration files. Find the Bit file we just generated and click Open. And choose not to attach SPI or

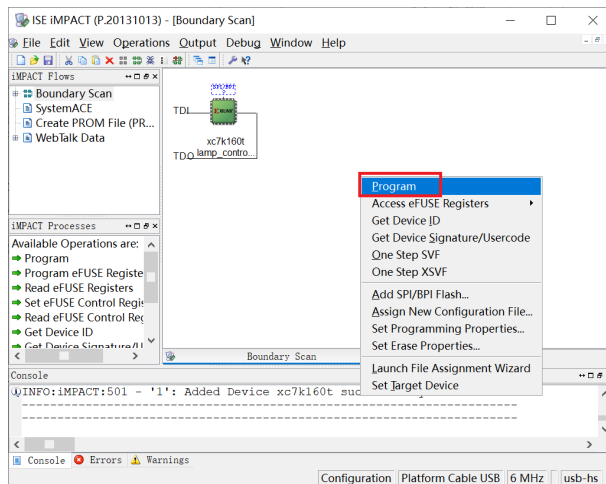
BPI PROM.



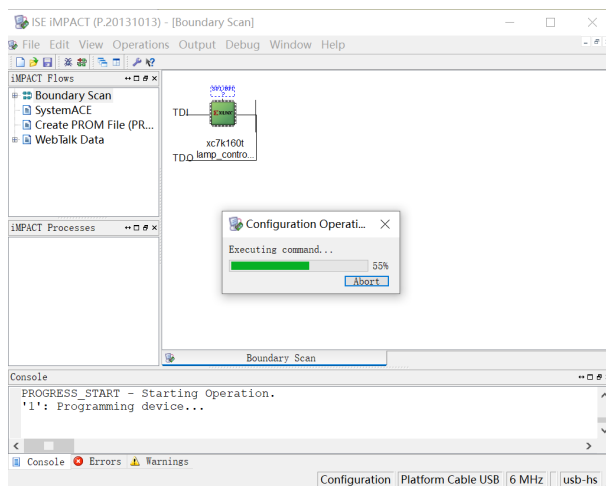
- Click OK at the Device Programming Properties window.



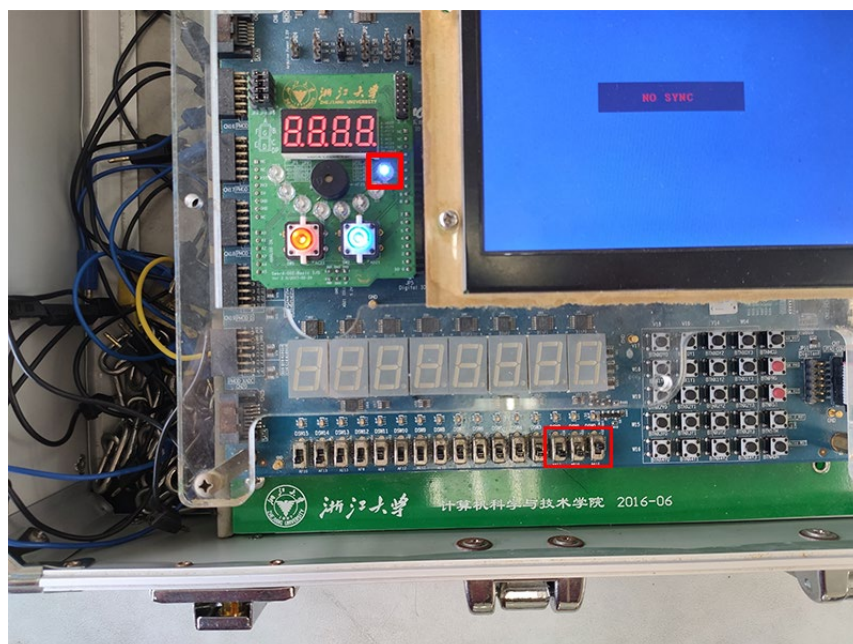
- Right click at the blank part of the window and click Program.



Wait for the program to be downloaded.



When it finishes, the program has been downloaded onto the board. We can test it using the switches AA10, AB10 and AA13. The corresponding LED is AF24.



5. Discussion and Revision of the Experiment

This is the first experiment that we combine what we have done on software with hardware. It's rather marvelous and rewarding to see our code is running with the right logical relationship on the SWORD board, though we didn't write the code from cover to cover. Besides, this experiment is considered as an introduction to ISE platform and Verilog language. It's the basis of the following series of experiments and the final project and thus is of vital significance.

Dealing with HDL and hardware is rather interesting in my point of view, and I'm waiting for doing more sophisticated projects with them.